Uncovering Singularity Signatures Behind Harmonic Spectra: A Complex-Analytic Framework for Gain-Invariant Timbre Representation

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Abstract

Timbre depends on the instantaneous spectrum of a sound and on the evolution of its harmonic amplitudes and phases. In this paper, we introduce dynamic complex singularity signatures, which present a compact, phase-aware generative model for the harmonic spectrotemporal component of quasi-harmonic sounds. At each time, finitely many algebraic singularity atoms, parameterised by depth, angular position, type, and complex amplitude, generate the full harmonic coefficient sequence. By letting the atom parameters evolve in time, we obtain the dynamic law, but keeping constant trajectories gives the static model. We derive the generating function and its singular data, prove identifiability up to permutation under minimality, formulate variable projection recovery with an integer-type matrix pencil initializer, and obtain a gauge-fixed local dynamic reconstruction theorem away from collisions. The same coefficient map gives closed-form brightness descriptors, exact mixture energy and centroid formulas including interference, and a pullback metric interpretation of spectral flux. Synthetic experiments provide validation, as recovery is exact to machine precision in the noiseless separated case with held-out complex NRMSE $\leq 4.2 \times 10^{-12}$. As predicted by the conditioning analysis, this degrades when components collide. Moreover, a signature fitted to 24 coefficients predicts the next 24 unseen coefficients with a median relative error of 1.5%, compared with a zero-predictor NRMSE of 1. For 72 isolated Iowa tones across two disjoint panels, it retains 93.6% of harmonic coefficient energy while storing 170 real values in place of 1088. Here, it demonstrates a median complex NRMSE of 0.253 and the harmonic-path SNR of 12.75 dB.

Keywords: Singularity analysis, generating functions, Fourier coefficients, harmonic spectra, inverse problems, identifiability, variable projection, timbre.

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1 Introduction

Timbre is the quality of sound that makes voices or musical instruments sound different from each other; that is, the characteristic that differentiates two sounds of the same pitch, loudness, and duration. Therefore, timbre acts as a defining characteristic of different musical instruments and is essential for instrument classification, audio synthesis, and sound reconstruction. The Fourier series plays an important role in the modelling of timbre. The Fourier coefficients of a periodic sound describe the amplitudes and phases of its harmonics, and the distribution and temporal evolution of harmonic energy are important components of timbre. However, representing a sound via a collection of a large number of Fourier coefficients is inefficient and encodes little about the explicit structure that relates one harmonic to another, while also failing to describe the evolution of the spectrum over time. This gap motivates a lower-dimensional generative model that captures patterns across harmonics and advances analysis, prediction, and reconstruction.

We aim to use harmonic-index singularities to create a signature storing a small number of essential parameters that capture underlying patterns across coefficients and can be used directly to model a part of the timbre. Specifically, they establish a phase-coherent harmonic component of a quasi-harmonic sound.

This singularity signature represents a harmonic coefficient as follows:

$$a_n \;=\; \sum_{k=1}^{K} c_k\, b_n(\theta_k)\, e^{-\tau_k n}\, e^{-in\varphi_k}, \qquad n \geq 1. \tag{1.1}$$

Here,

$$b_n(\theta) \;=\; \frac{\Gamma(n+\theta)}{\Gamma(\theta)\,\Gamma(n+1)}, \tag{1.2}$$

with $\tau_k > 0$, $\theta_k > 0$, and $c_k \neq 0$.

However, a stationary spectrum is insufficient to accurately model the evolving character of a note. Timbre is continuously dependent on attacks, brightness changes, and differential harmonic decay, among other factors. Therefore, we establish a dynamic law:

$$a_n(t) \;=\; \sum_{k=1}^{K} c_k(t)\, b_n(\theta_k(t))\, e^{-\tau_k(t)\, n}\, e^{-in\varphi_k(t)}. \tag{1.3}$$

The singularity signatures obtained, where K determines the number of atoms in each signature, are given as:

$$\Sigma(t) \;=\; \Big\{\big(\tau_k(t),\ \varphi_k(t),\ \theta_k(t),\ c_k(t)\big)\Big\}_{k=1}^{K}. \tag{1.4}$$

Section 2 covers basic preliminaries regarding sound wave representation and the Fourier series. Next, Section 3 develops the static singularity-signature model. We intro-

duce a complex variable z and construct a generating function for the Fourier coefficients by summing equation (1.1) against powers of z, obtaining a closed form whose singularities occur at $z_k = e^{\tau_k + i\varphi_k}$. With the signature established, we place it in a permutation quotient space to remove component ordering, and in a gain and time origin quotient space to compare only timbre. Section 4 develops the inverse problem at fixed time to establish recoverability. This includes using variable projection to separate the linear and nonlinear singularity parameters and, for integer values of θ, a recurrence-based initialiser for the nonlinear coordinates. We also use an exact Jacobian for analysing identifiability and conditioning, while estimator variance is evaluated using Fisher information under a Gaussian noise model. Section 5 extends the model to the dynamic law. We fix the carrier phase externally to make the time-varying coefficients $a_n(t)$ identifiable, use a windowed analysis convention, and continuously track components away from collisions, obtaining a reconstruction theorem. Section 6 uses the model to represent classical descriptors of timbre, including brightness, tilt, envelope peak, attack and decay, coherent modulation, and spectral flux. Section 7 develops practical applications of the model, including harmonic analysis and resynthesis, prediction of unseen Fourier coefficients, and descriptor-based comparison and classification. Finally, in Section 8, we perform empirical testing for sound reconstruction, instrument classification, and Fourier coefficient recovery on synthetic and real-world datasets.

Existing spectral models use the harmonics themselves as the baseline representation of timbre. While sinusoidal analysis-synthesis stores the phase and amplitude for every harmonic in each frame [13], spectral modelling synthesis is used to add a stochastic residual [16]. Both methods accurately reconstruct sound, but neither relates the harmonic n to harmonic $n+1$. Therefore, the parameter count increases with the number of harmonics retained. On the other hand, descriptor methods reduce a spectrum to scalars such as the spectral centroid [15], but these discard phase and cannot resynthesise. Further, perceptual studies show that the attack carries much of a listener's discrimination between instruments [7], so a compact, phase-aware, time-varying representation is required. Our contributions include the development of a singularity-based representation of harmonic coefficients, a fixed-time recovery framework with identifiability and conditioning analysis, and a dynamic extension for reconstruction error, phase, and windowing. Furthermore, we show that the singularity parameters established present mathematical interpretations of timbre-related quantities, including brightness, spectral tilt, and attack. Hence, they connect the complex-analytic model to measurable acoustic behaviour.

2 Preliminaries

2.1 Audio signals and Fourier analysis

Definition 2.1 (Audio signal)**.** An *audio signal* is a function $x\colon \mathbb{R} \to \mathbb{R}$, where $x(t)$ gives the amplitude of the sound at time t seconds. We assume that, for every bounded interval $I \subset \mathbb{R}$, $x \in L^2_{loc}(\mathbb{R})$, that is, $\int_I |x(t)|^2\,dt < \infty$.

Definition 2.2 (Pure tone and periodic sound)**.** We define a pure tone to be a single-frequency sound:

$$x(t) = A\cos(2\pi f_0 t + \varphi)\,, \tag{2.1}$$

where $A \geq 0$ is the amplitude and $f_0 > 0$ the fundamental frequency. A signal is said to be periodic if $x(t+T) = x(t)$ for all t, with period $T = \frac{1}{f_0}$. Throughout, $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ denotes the circle, so that a function on $\mathbb{T}$ is one of period 2π in the normalised phase variable.

Theorem 2.3 (Fourier series [10])**.** *Every square-integrable periodic function g can be expanded as a sum of complex harmonics, with convergence in $L^2(\mathbb{T})$:*

$$g(u) \;=\; \sum_{n=-\infty}^{\infty} \hat{g}_n\, e^{inu}, \qquad \hat{g}_n \;=\; \frac{1}{2\pi}\int_{\mathbb{T}} g(u)\, e^{-inu}\,du. \tag{2.2}$$

According to Parseval's theorem, the total energy of the signal is equal to the total energy of its Fourier coefficients:

$$\frac{1}{2\pi}\int_{\mathbb{T}} |g(u)|^2\,du \;=\; \sum_{n=-\infty}^{\infty} \left|\hat{g}_n\right|^2. \tag{2.3}$$

The complex exponentials $\{e^{inu}\}_{n\in\mathbb{Z}}$ are orthonormal under

$$\langle g, h\rangle = (2\pi)^{-1}\int_{\mathbb{T}} g\,\overline{h}\,du, \qquad \left\langle e^{imu}, e^{inu}\right\rangle = \delta_{mn}. \tag{2.4}$$

Lemma 2.4 (One-sided representation of a real periodic sound)**.** *For a square-integrable function $g\colon \mathbb{T} \to \mathbb{R}$ with Fourier coefficients $\hat{g}_n$:*

$$\hat{g}_{-n} = \overline{\hat{g}_n}, \qquad g(u) \;=\; \hat{g}_0 + 2\,\mathrm{Re}\sum_{n=1}^{\infty} \hat{g}_n e^{inu}. \tag{2.5}$$

Proof. Since the function g is real, $\overline{g(u)} = g(u)$, and by (2.2), $\hat{g}_{-n} = \frac{1}{2\pi}\int_{\mathbb{T}} g(u)\,e^{inu}\,du = \overline{\frac{1}{2\pi}\int_{\mathbb{T}} g(u)\,e^{-inu}\,du} \;=\; \overline{\hat{g}_n}$. Splitting the expansion (2.2) into positive and negative frequencies gives $g(u) = \hat{g}_0 + \sum_{n=1}^{\infty}\hat{g}_n e^{inu} + \sum_{n=1}^{\infty}\hat{g}_{-n}e^{-inu} = \hat{g}_0 + \sum_{n=1}^{\infty}\Big(\hat{g}_n e^{inu} + \overline{\hat{g}_n e^{inu}}\Big)$. Substituting $\hat{g}_{-n} = \overline{\hat{g}_n}$ gives (2.5). ∎

2.2 Locally harmonic signals and the frozen cycle

Definition 2.5 (Carrier phase and harmonic channel)**.** For a real signal $x(t)$ that is locally harmonic on an interval I, we fix a carrier phase $\Phi\colon I \to \mathbb{R}$ satisfying:

$$\Phi'(t) = 2\pi f_0(t) > 0, \tag{2.6}$$

where $f_0\colon I \to (0,\infty)$ is the instantaneous fundamental frequency. This extends the constant f_0 given in Definition 2.2. The *harmonic part* of the signal is then

$$x_H(t) \;=\; a_0(t) + 2\,\mathrm{Re}\sum_{n=1}^{\infty} a_n(t)\, e^{in\Phi(t)}. \tag{2.7}$$

The remaining part $x_R(t) = x(t) - x_H(t)$ is not included in the model, since it contains unmodelled noise, inharmonicity, reverberation, and analysis error.

Definition 2.6 (Frozen-cycle waveform)**.** For each fixed t, the *frozen-cycle waveform* is given by

$$g_t(u) \;=\; a_0(t) + 2\,\mathrm{Re}\sum_{n=1}^{\infty} a_n(t)\, e^{inu}, \qquad u \in \mathbb{T}. \tag{2.8}$$

Lemma 2.7 (Evaluation and per-frame Fourier identities)**.** *Suppose that the coefficient sequence $\{a_n(t)\}_{n\geq 1}$ is absolutely summable for each fixed t. Then:*

$$\text{(i)}\quad x_H(t) = g_t(\Phi(t)), \tag{2.9}$$

and for every $n \geq 1$:

$$\text{(ii)}\quad a_n(t) = \frac{1}{2\pi}\int_{\mathbb{T}} g_t(u)\, e^{-inu}\, du, \qquad a_0(t) = \frac{1}{2\pi}\int_{\mathbb{T}} g_t(u)\, du. \tag{2.10}$$

Proof. For (i), substitute $u = \Phi(t)$ into (2.8) and compare with (2.7).

For (ii), write the frozen cycle function g_t by Lemma 2.4 in its two-sided Fourier series $g_t(u) = \sum_{m\in\mathbb{Z}} \hat{g}_m e^{imu}$, where $\hat{g}_0 = a_0(t)$, $\hat{g}_m = a_m(t)$ for $m \geq 1$, and $\hat{g}_m = \overline{a_{-m}(t)}$ for $m \leq -1$. Since the coefficients are absolutely summable, the Fourier series can be integrated term by term against the bounded function e^{-inu}. The orthonormality relations (2.4) give $\frac{1}{2\pi}\int_{\mathbb{T}} g_t(u)\, e^{-inu}\, du = \sum_{m\in\mathbb{Z}} \hat{g}_m \cdot \frac{1}{2\pi}\int_{\mathbb{T}} e^{i(m-n)u}\, du = \sum_{m\in\mathbb{Z}} \hat{g}_m \delta_{mn} = \hat{g}_n$. This is equal to $a_n(t)$ for $n \geq 1$ and to $a_0(t)$ for $n = 0$. ∎

All coefficient-domain results are therefore applied to g_t, since it captures the instantaneous harmonic shape.

2.3 The gamma-envelope function

Definition 2.8 (Atom envelope function)**.** For $\theta > 0$, we define the *atom envelope function*

$$b_n(\theta) \;=\; \frac{\Gamma(n+\theta)}{\Gamma(\theta)\,\Gamma(n+1)} \;=\; \frac{(\theta)_n}{n!}, \qquad n \geq 0, \tag{2.11}$$

where Γ is the gamma function and $(\theta)_n = \theta(\theta+1)\cdots(\theta+n-1)$ is the Pochhammer symbol.

Lemma 2.9 (Elementary properties of b_n)**.** *For every $\theta > 0$ and $n \geq 0$: (i) $b_0(\theta) = 1$ and $b_n(\theta) > 0$; (ii) both the expressions in (2.11) agree; (iii) the following ratio recurrence holds:*

$$\frac{b_{n+1}(\theta)}{b_n(\theta)} \;=\; \frac{n+\theta}{n+1}. \tag{2.12}$$

Proof. (i) Since $\Gamma(1) = 1$, $b_0(\theta) = \frac{\Gamma(\theta)}{\Gamma(\theta)\,\Gamma(1)} = 1$. Here, Γ is strictly positive on $(0,\infty)$, so every factor in $\frac{\Gamma(n+\theta)}{\Gamma(\theta)\,\Gamma(n+1)}$ is positive and therefore positivity follows.

(ii) Repeatedly using $\Gamma(z+1) = z\,\Gamma(z)$ [14] gives $\Gamma(n+\theta) = \theta(\theta+1)\cdots(\theta+n-1)\,\Gamma(\theta)$, and since $\Gamma(n+1) = n!$, we get the second expression.

(iii) Using $\Gamma(z+1) = z\,\Gamma(z)$ both in the numerator and in the denominator, we get $\frac{b_{n+1}(\theta)}{b_n(\theta)} = \frac{\Gamma(n+1+\theta)}{\Gamma(n+2)} \cdot \frac{\Gamma(n+1)}{\Gamma(n+\theta)} = \frac{(n+\theta)\,\Gamma(n+\theta)}{(n+1)\,\Gamma(n+1)} \cdot \frac{\Gamma(n+1)}{\Gamma(n+\theta)} = \frac{n+\theta}{n+1}$.

■

Theorem 2.10 (Generalised binomial series [14])**.** *For $\theta > 0$ and $|y| < 1$,*

$$\sum_{n=0}^{\infty} b_n(\theta)\, y^n \;=\; (1-y)^{-\theta}. \tag{2.13}$$

Proposition 2.11 (Gamma-ratio asymptotic [14])**.** *For fixed θ, as $n \to \infty$,*

$$b_n(\theta) \;=\; \frac{n^{\theta-1}}{\Gamma(\theta)}\Big[1 + \frac{\theta(\theta-1)}{2n} + O\big(n^{-2}\big)\Big]. \tag{2.14}$$

3 The Static Singularity-Signature Model

This section defines and derives the singularity signature along with the coefficient law. We also establish the complex-analytic structure, including the closed form of the generating function, its singular data, the asymptotic envelope, and the quotient spaces that are used to compare signatures.

3.1 Singularity atoms and the static coefficient law

Definition 3.1 (Singularity atom and static coefficient law)**.** We define, for each component k,

$$w_k = e^{-\tau_k - i\varphi_k}. \tag{3.1}$$

Since $\tau_k > 0$, $\left|w_k\right| = e^{-\tau_k} < 1$. The *static coefficient law* of a K-component signature, for $n \geq 1$, is

$$a_n \;=\; A_n(\Sigma) \;=\; \sum_{k=1}^{K} c_k\, b_n(\theta_k)\, w_k^n. \tag{3.2}$$

Here, $A_n \colon S_K \to \mathbb{C}$ denotes the coefficient map at harmonic index n; we also write $A_N = \big(A_1, \ldots, A_N\big)$ for the corresponding map into $\mathbb{C}^N$.

The *singularity signature* is:

$$\Sigma \;=\; \Big\{\big(\tau_k, \varphi_k, \theta_k, c_k\big)\Big\}_{k=1}^{K}, \tag{3.3}$$

where τ_k is the singularity depth, governing exponential attenuation of the high harmonics; φ_k the angular position in period phase; θ_k the singularity type, governing the algebraic curvature of the envelope; and c_k the complex strength and common phase of the component.

3.2 The generating function and its singular data

Definition 3.2 (Generating function of the coefficient law)**.** Assuming $\tau_k \geq \tau_{min} > 0$ and $\theta_k > 0$, we define the following generating function for Fourier coefficients, with the value of a_n given by (3.2):

$$F_\Sigma(z) = \sum_{n=1}^{\infty} a_n z^n. \tag{3.4}$$

Theorem 3.3 (Closed form of the generating function)**.** *F_Σ, given in Definition 3.2, is analytic for $|z| < \min_k \left|w_k\right|^{-1} = e^{\tau_{min}}$, and*

$$F_\Sigma(z) \;=\; \sum_{k=1}^{K} c_k \Big[\big(1 - w_k z\big)^{-\theta_k} - 1\Big]. \tag{3.5}$$

Proof. We fix z for $|z| < e^{\tau_{min}}$, and therefore $\left|w_k z\right| \leq e^{-\tau_k}|z| < 1$ holds for every k. Substituting (3.2) gives $F_\Sigma(z) = \sum_{n=1}^{\infty} \sum_{k=1}^{K} c_k\, b_n(\theta_k) \left(w_k z\right)^n$. This series converges absolutely, and on applying Theorem 2.10 to $y = \left|w_k z\right| < 1$, we get $\sum_{n=1}^{\infty} \sum_{k=1}^{K} \Big|c_k\, b_n(\theta_k) \left(w_k z\right)^n\Big| = \sum_{k=1}^{K} \left|c_k\right| \sum_{n=1}^{\infty} b_n(\theta_k) \left|w_k z\right|^n \;\leq\; \sum_{k=1}^{K} \left|c_k\right| \Big(1 - \left|w_k z\right|\Big)^{-\theta_k} \;<\; \infty$, using $b_n(\theta_k) > 0$ by Lemma 2.9. The finite component sum and infinite series can be interchanged because of absolute convergence: $F_\Sigma(z) \;=\; \sum_{k=1}^{K} c_k \sum_{n=1}^{\infty} b_n(\theta_k) \left(w_k z\right)^n$. Adding and subtract-

ing the term $b_0(\theta_k) = 1$ from Lemma 2.9(i) inside each inner series gives $F_\Sigma(z) = \sum_{k=1}^{K} c_k \Big[\sum_{n=0}^{\infty} b_n(\theta_k) \big(w_k z\big)^n - 1\Big]$. Setting $y = w_k z$ and applying the generalised binomial series (2.13) gives (3.5). Each term is analytic for $\big|w_k z\big| < 1$, and therefore F_Σ is analytic for $|z| < e^{\tau_{min}}$. ■

Corollary 3.4 (Singular data [3]). *Under Theorem 3.3, assuming the points $z_1, \ldots, z_K$ to be pairwise distinct, the closed form (3.5) of F_Σ extends analytically to*

$$\mathbb{C} \setminus \bigcup_k R_k.$$

Here, $R_k = \big\{z\colon 1 - w_k z \in (-\infty, 0]\big\}$, with each factor $\big(1 - w_k z\big)^{-\theta_k}$ taken in its principal branch. Its finite singularities are located exactly at

$$z_k \;=\; w_k^{-1} \;=\; e^{\tau_k + i\varphi_k}, \qquad k = 1, \ldots, K. \tag{3.6}$$

Near z_k, the k-th component behaves as $c_k\big(1 - \frac{z}{z_k}\big)^{-\theta_k}$. Therefore, it acts like a branch point for non-integer θ_k and a pole of order θ_k for integer θ_k.

Proof. For each k, the principal branch of $\big(1 - w_k z\big)^{-\theta_k}$ is analytic wherever $1 - w_k z \notin (-\infty, 0]$. Therefore, it is analytic on $\mathbb{C} \setminus R_k$. Since (3.5) is a finite sum, F_Σ is analytic on

$$\mathbb{C} \setminus \bigcup_{k=1}^{K} R_k.$$

Now, because $w_k = z_k^{-1}$, we have $1 - w_k z = 1 - \frac{z}{z_k}$. Thus the k-th term is singular at $z = z_k$ and behaves there as $c_k\big(1 - \frac{z}{z_k}\big)^{-\theta_k}$. All the terms that have different singularity locations are analytic near z_k, which gives the local singular behaviour. ■

3.3 Asymptotic envelope law

Proposition 3.5 (Isolated-component envelope [3]). *Given $w = e^{-\tau - i\varphi}$, $\tau > 0$, an isolated component $a_n = c\, b_n(\theta)\, w^n$ satisfies:*

$$\big|a_n\big| \;=\; \frac{|c|}{\Gamma(\theta)}\, n^{\theta - 1}\, e^{-\tau n} \Big[1 + O\big(n^{-1}\big)\Big]. \tag{3.7}$$

Proof. Take absolute values of the component a_n: $\big|a_n\big| = |c|\, b_n(\theta)\, |w|^n \big|e^{-in\varphi}\big|$. Since $|w| = e^{-\tau}$, $\big|a_n\big| = |c|\, b_n(\theta)\, e^{-\tau n}$. Now, substituting the expansion (2.14) for $b_n(\theta)$ gives (3.7). ■

Remark: (3.7) shows how τ controls the exponential decay at high harmonic numbers, while θ provides an algebraic correction along with a low-to-mid-harmonic curvature.

3.4 Signature space and the timbre-comparison quotient

Definition 3.6 (Signature space)**.** Let the single-atom parameter space (depth, angle, type, complex amplitude) be represented by $H = (0,\infty) \times \mathbb{T} \times \Theta \times \mathbb{C}^*$, where $\Theta = (0,\infty)$ is the admissible set of singularity types, and let $\Delta \subset H^K$ represent the collision set:

$$\Delta \;=\; \Big\{ \big(h_1,\dots,h_K\big) \in H^K \colon \big(\tau_i,\varphi_i,\theta_i\big) = \big(\tau_j,\varphi_j,\theta_j\big) \;\; for\ some\ i \neq j \Big\}. \tag{3.8}$$

The *unordered signature space* is the quotient

$$S_K \;=\; \big(H^K \setminus \Delta\big)/\mathfrak{S}_K, \tag{3.9}$$

where $\mathfrak{S}_K$ is the symmetric group acting by permuting the K components.

Definition 3.7 (Timbre-comparison quotient)**.** For the physical reconstruction of the actual sound, the phase origin and the amplitude c must be retained. For timbre comparison alone, however, positive gain and time-origin shift may be ignored by identifying:

$$c_k \mapsto g\, c_k, \quad g > 0; \qquad \varphi_k \mapsto \varphi_k + \gamma. \tag{3.10}$$

We write $S_K^{tim} = S_K/\sim$ for the resulting timbre-comparison quotient, where $\sim$ is the equivalence relation on S_K that is generated by (3.10).

4 Recovery, Identifiability, and Conditioning

This section covers singularity signature recovery based on observed Fourier coefficients at a fixed time. It develops the variable-projection estimator and a recurrence-based initialiser for integer types. We also derive the exact Jacobian, Fisher information, and establish identifiability.

4.1 Observation model and variable projection

Definition 4.1 (Observation model)**.** Let $\xi = \big(\log\tau_k, \varphi_k, \theta_k\big)_{k=1}^K$ collect the nonlinear atom parameters of the signature. Let $n_1 < n_2 < \cdots < n_N$ denote the observed harmonic indices. We define the *design matrix* $G(\xi) \in \mathbb{C}^{N\times K}$, where N is the number of harmonic indices, entrywise by

$$G_{jk}(\xi) \;=\; b_{n_j}(\theta_k)\, e^{-\tau_k n_j}\, e^{-i n_j \varphi_k}. \tag{4.1}$$

The static law (3.2) restricted to the observed indices then reads

$$a \;=\; G(\xi)\, c \;+\; \varepsilon. \tag{4.2}$$

Here $c = \big(c_1, \ldots, c_K\big)^T$, and ε collects measurement and model error.

Definition 4.2 (Separable objective)**.** Let $\mathcal{W}$ be a positive diagonal weight matrix. Then, the separable least-squares objective is given by

$$Q(\xi, c) \;=\; \|\mathcal{W}^{\frac{1}{2}}\big(a - G(\xi)\, c\big)\|_2^2. \tag{4.3}$$

Lemma 4.3 (Conditional minimiser in c)**.** *Set $\tilde{a} = \mathcal{W}^{\frac{1}{2}}a$ and $\tilde{G} = \mathcal{W}^{\frac{1}{2}}G(\xi)$. Every minimiser of $Q(\xi, c)$ then satisfies the normal equations:*

$$\tilde{G}^*\tilde{G}\, c \;=\; \tilde{G}^*\tilde{a}. \tag{4.4}$$

The minimum-norm solution, using the Moore–Penrose pseudoinverse where needed [6], is given by:

$$c^\dagger(\xi) \;=\; \tilde{G}(\xi)^\dagger\tilde{a} \;=\; \big(G^*\mathcal{W}G\big)^\dagger G^*\mathcal{W}a. \tag{4.5}$$

Proof. Expanding the objective given in (4.3) as a Hermitian quadratic form in c yields $Q(\xi, c) \;=\; \big(\tilde{a} - \tilde{G}c\big)^*\big(\tilde{a} - \tilde{G}c\big) \;=\; \tilde{a}^*\tilde{a} - c^*\tilde{G}^*\tilde{a} - \tilde{a}^*\tilde{G}c + c^*\tilde{G}^*\tilde{G}c$. We treat c and $\bar{c}$ as independent variables in the Wirtinger sense, which is standard for complex least-squares equations [11]. Differentiating with respect to $\bar{c}$ and setting the derivative to zero, we get $\frac{\partial Q}{\partial \bar{c}} = -\tilde{G}^*\tilde{a} + \tilde{G}^*\tilde{G}c = 0$. This is precisely (4.4). The stationary point is the global minimum of the convex quadratic because $\tilde{G}^*\tilde{G}$ is positive semidefinite. Moreover, by the standard pseudoinverse identity $A^\dagger = \big(A^*A\big)^\dagger A^*$ [6], the minimum-norm solution of (4.4) is given by $c = \big(\tilde{G}^*\tilde{G}\big)^\dagger\tilde{G}^*\tilde{a} = \tilde{G}^\dagger\tilde{a}$. We get the right-hand-side expression of (4.5) by unfolding $\tilde{G} = \mathcal{W}^{\frac{1}{2}}G$ and $\tilde{a} = \mathcal{W}^{\frac{1}{2}}a$. ■

Theorem 4.4 (Variable projection [5])**.** *With the orthogonal projector $P_{\tilde{G}} = \tilde{G}\tilde{G}^\dagger$, substituting the optimal amplitude vector from (4.5) into the least-squares objective gives*

$$Q_{prof}(\xi) \;=\; \|\big(I - P_{\tilde{G}(\xi)}\big)\tilde{a}\|_2^2. \tag{4.6}$$

Proof. Since $c = c^\dagger(\xi) = \tilde{G}^\dagger\tilde{a}$, the residual is $\tilde{a} - \tilde{G}c^\dagger = \tilde{a} - \tilde{G}\tilde{G}^\dagger\tilde{a} = \big(I - P_{\tilde{G}}\big)\tilde{a}$. Therefore, $Q\big(\xi, c^\dagger(\xi)\big) = \|\big(I - P_{\tilde{G}}\big)\tilde{a}\|_2^2$. ■

Hence, rather than searching over $5K$ real parameters, the variable projection had to search over only $3K$ nonlinear parameters ($\log\tau_k, \varphi_k, \theta_k$ per component), with the $2K$ real amplitude parameters obtained in closed form at each step.

Let $B \subset \mathbb{R}^{3K}$ denote the admissible parameter set, which enforces $\theta_k > 0$, $\tau_k > 0$, and angular wrapping of φ_k, along with any predeclared upper bounds. The framewise estimate is

$$\hat{\xi} \;=\; \operatorname*{arg\,min}_{\xi \in B}\; Q_{prof}(\xi), \qquad \hat{c} \;=\; c^\dagger\big(\hat{\xi}\big). \tag{4.7}$$

4.2 Integer-type recurrences and the matrix-pencil initialiser

Lemma 4.5 (Integer-type atoms are exponential polynomials [17])**.** *Substituting $\theta_k = m_k$ into $b_n(\theta_k)$ gives*

$$b_n(m_k) \;=\; \binom{n+m_k-1}{m_k-1}. \tag{4.8}$$

This is a polynomial in n of degree m_k-1. As a result, the k-th component can be written as

$$c_k\, b_n(m_k)\, w_k^n \;=\; p_k(n)\, \lambda_k^n. \tag{4.9}$$

Therefore, each integer-type component is an exponential polynomial $p_k(n)\,\lambda_k^n$ where $\lambda_k = w_k$ and $\deg p_k = m_k - 1$.

Proof. For integer m_k, by Definition 2.8:

$$b_n(m_k) = \frac{\Gamma(n+m_k)}{\Gamma(m_k)\,\Gamma(n+1)} = \frac{(n+m_k-1)!}{(m_k-1)!\;n!} = \binom{n+m_k-1}{m_k-1}.$$

Also, the binomial coefficient can be written in product form,

$$\binom{n+m_k-1}{m_k-1} = \frac{(n+1)(n+2)\cdots(n+m_k-1)}{(m_k-1)!}.$$

Therefore, the component contribution to a_n is $c_k\, b_n(m_k)\, w_k^n = p_k(n)\,\lambda_k^n$ with $p_k(n) = c_k\, b_n(m_k)$ and $\lambda_k = w_k$. ∎

Proposition 4.6 (Annihilation identity [17])**.** *Let E be the forward shift operator, $(Ea)_n = a_{n+1}$. Then, for $\lambda \in \mathbb{C}\setminus\{0\}$ and for every polynomial p of degree at most $m-1$,*

$$(E-\lambda)^m\big[p(n)\,\lambda^n\big] \;=\; 0. \tag{4.10}$$

Proof. For $m=1$, $(E-\lambda)\big[p(n)\,\lambda^n\big] = \lambda^{n+1}\big(p(n+1)-p(n)\big)$, and $p(n+1)-p(n)$ has degree at most $\deg p - 1$. Iterating m times lowers the degree below zero and gives the zero sequence. ∎

Proposition 4.7 (Complete annihilating recurrence [17])**.** *Let the nodes $\lambda_k = w_k$ be distinct and suppose $\theta_k = m_k$ are positive integers. Then, the sequence $\{a_n\}$ of (3.2) is annihilated by*

$$P(E) \;=\; \prod_{k=1}^{K}\big(E-\lambda_k\big)^{m_k} \;=\; \sum_{j=0}^{r} p_j E^j, \qquad r \;=\; \sum_{k=1}^{K} m_k. \tag{4.11}$$

Hence, for every valid n,

$$\sum_{j=0}^{r} p_j\, a_{n+j} \;=\; 0. \tag{4.12}$$

Proof. By Lemma 4.5, $a_n = \sum_k p_k(n)\,\lambda_k^n$. By Proposition 4.6, for each k:

$$\big(E - \lambda_k\big)^{m_k}\big[p_k(n)\,\lambda_k^n\big] = 0.$$

Here, all the factors of $P(E)$ are polynomials in the same shift operator, E, and they commute:

$$P(E) = \Big[\prod_{l\neq k}\big(E - \lambda_l\big)^{m_l}\Big]\big(E - \lambda_k\big)^{m_k}.$$

Therefore, each component $p_k(n)\,\lambda_k^n$ is annihilated by $P(E)$. By linearity:

$$P(E)\big[a_n\big] = \sum_k P(E)\big[p_k(n)\,\lambda_k^n\big] = 0.$$

Now, expanding $P(E) = \sum_{j=0}^{r} p_j E^j$ yields (4.12). ∎

Definition 4.8 (Hankel recurrence system [9])**.** First, normalise $p_r = 1$. Then, for row index $l = 1, \ldots, N - r$ and column index $j = 1, \ldots, r$, we define the Hankel recurrence matrix, unknown vector, and right-hand side by

$$\big(H_r\big)_{lj} = a_{l+j-1}, \qquad p = \big(p_0, p_1, \ldots, p_{r-1}\big)^T, \qquad b = \big(a_{r+1}, a_{r+2}, \ldots, a_N\big)^T. \tag{4.13}$$

Corollary 4.9 (Linear recurrence system [9])**.** *By Definition 4.8, the annihilation relation (4.12) is equivalent to the linear system:*

$$H_r\,p \;=\; -\,b. \tag{4.14}$$

The system's l-th row is given by

$$a_l\,p_0 + a_{l+1}\,p_1 + \cdots + a_{l+r-1}\,p_{r-1} \;=\; -\,a_{l+r}, \qquad l = 1, \ldots, N - r. \tag{4.15}$$

Proof. Set $n = l$ in (4.12): $\sum_{j=0}^{r} p_j\,a_{l+j} = 0$. Since $p_r = 1$, isolating the $j = r$ term gives $\sum_{j=0}^{r-1} p_j\,a_{l+j} = -\,a_{l+r}$. This is (4.15). Now, stacking all the rows gives the exact Hankel entries $\big(H_r\big)_{lj} = a_{l+j-1}$ and right side $-\,b$. ∎

Remark: At least $2r$ consecutive coefficients are needed for a square noiseless system. With more coefficients available, a least-squares solution is preferred. The roots of $p_0 + p_1 z + \cdots + z^r$ estimate the nodes λ_k, repeated according to multiplicities m_k. They can be mapped to physical parameters by

$$\tau_k \;=\; -\log\big|\lambda_k\big|, \qquad \varphi_k \;=\; -\arg\lambda_k \;(\mathrm{mod}\;2\pi). \tag{4.16}$$

For simple exponentials, the same nodes can be recovered with a Hankel pencil matrix where $H_0 = \big(a_{i+j}\big)$ and $H_1 = \big(a_{i+j+1}\big)$. Here, the generalised eigenvalues of $H_1 v = \lambda H_0 v$

are the nodes [9]. For $m_k > 1$, repeated nodes become unstable under noise and create a confluent problem. Therefore, pencil recovery should be used as an initialiser before variable-projection refinement.

Definition 4.10 (Minimal signature)**.** A signature $\Sigma = \{(\tau_k, \varphi_k, \theta_k, c_k)\}_{k=1}^{K}$ is *minimal* if $c_k \neq 0$ for every k and no pair (w_k, θ_k) is duplicated, that is, $(w_i, \theta_i) \neq (w_j, \theta_j)$ whenever $i \neq j$.

Proposition 4.11 (No finite recurrence for fractional type)**.** *If θ_k is not a positive integer, and the signature is minimal (Definition 4.10) and node-distinct, then there is no finite-order linear recurrence with constant coefficients that can annihilate the sequence. Therefore, the full nonlinear model must be used.*

Proof. A complex sequence satisfies an annihilating recurrence only if its generating function is rational [17]. By Theorem 3.3 and Corollary 3.4, the generating function F_Σ has a branch-point singularity at the point $z_k = w_k^{-1}$. Since a rational function can only have poles and not branch points, F_Σ is not rational and no finite recurrence exists. ■

4.3 Jacobian and identifiability

Proposition 4.12 (Componentwise derivatives)**.** *For one component, we define*

$$g_n \;=\; b_n(\theta)\, e^{-\tau n}\, e^{-in\varphi}, \qquad q_n \;=\; c\, g_n. \tag{4.17}$$

Here, $c = u + iv$ and $\alpha = \log \tau$. Then,

$$\frac{\partial q_n}{\partial \alpha} = -\,\tau\, n\, q_n, \qquad \frac{\partial q_n}{\partial \varphi} = -\,i\, n\, q_n, \qquad \frac{\partial q_n}{\partial u} = g_n, \qquad \frac{\partial q_n}{\partial v} = i\, g_n. \tag{4.18}$$

Moreover,

$$\frac{\partial q_n}{\partial \theta} \;=\; q_n\big[\psi(n+\theta) - \psi(\theta)\big]. \tag{4.19}$$

Here, $\psi = \frac{\Gamma'}{\Gamma}$ is the digamma function [14].

Proof. Write $q_n = c\, b_n(\theta)\, e^{-\tau n} e^{-in\varphi}$, $\tau = e^{\alpha}$. Then $\frac{\partial q_n}{\partial \tau} = -\,n\, q_n$ and $\frac{d\tau}{d\alpha} = \tau$, so by the chain rule, $\frac{\partial q_n}{\partial \alpha} = -\,\tau n\, q_n$. Also, $\frac{\partial q_n}{\partial \varphi} = -\,in\, q_n$. Since $q_n = (u + iv)\, g_n$ is linear in u and v, $\frac{\partial q_n}{\partial u} = g_n$ and $\frac{\partial q_n}{\partial v} = i\, g_n$. For θ, $\log b_n(\theta) = \log \Gamma(n+\theta) - \log \Gamma(\theta) - \log \Gamma(n+1)$, so $\frac{\partial}{\partial \theta} \log b_n(\theta) = \psi(n+\theta) - \psi(\theta)$. Since θ enters q_n only through $b_n(\theta)$, $\frac{\partial q_n}{\partial \theta} = q_n\big[\psi(n+\theta) - \psi(\theta)\big]$, which is (4.19). ■

Definition 4.13 (Realified Jacobian)**.** We define a complex Jacobian $J \in \mathbb{C}^{N \times 5K}$, and stack the five real-parameter columns (4.18)–(4.19) for every component into it. To work with real parameters, we convert it to

$$J_R \;=\; \big[\operatorname{Re} J \quad \operatorname{Im} J\big] \in \mathbb{R}^{2N \times 5K}. \tag{4.20}$$

Theorem 4.14 (Infinite-data identifiability)**.** *Let the singularity signature*

$$\Sigma = \{(\tau_k, \varphi_k, \theta_k, c_k)\}_{k=1}^{K}$$

be minimal (Definition 4.10), and assume all w_k to be distinct. Then, the coefficient sequence $\{a_n\}_{n\geq 1}$ determines the singularity locations w_k^{-1}, the exponents θ_k, and the coefficients c_k uniquely, up to permutation of components.

Proof. Step 1: the full sequence of coefficients $\{a_n\}$ is used to determine the generating function $F_\Sigma(z) = \sum_{n\geq 1} a_n z^n$. By Theorem 3.3, this is convergent on the disk $D = \{|z| < e^{\tau_{min}}\}$. By the identity theorem for analytic functions [1], any analytic function that agrees with F_Σ on a neighbourhood of 0 agrees on all of D. By Corollary 3.4, the continuation of F_Σ to the connected domain

$$\mathbb{C} \setminus \bigcup_k R_k$$

exists. Therefore, it is unique by the identity theorem.

Step 2: recover the singularity locations. By Corollary 3.4, the finite singularities are located at $\{z_j\} = \{w_k^{-1}\}$. At every one of these points, there is at least one term behaving like $c_k\big(1 - \frac{z}{z_k}\big)^{-\theta_k}$. This diverges because $\theta_k > 0$. Therefore, each z_j is a singularity. The singular set of the continuation is equal to $\{w_k^{-1}\}$, and each location is recovered. Moreover, by (3.6), $\tau_k = \log\big|z_k\big|$ and $\varphi_k = \arg z_k$.

Step 3: recover exponents and coefficients. First, fix a singular location z_j which has ordered exponents $\theta_{j,1} > \theta_{j,2} > \cdots$ and coefficients $c_{j,1}, c_{j,2}, \ldots$. Then, approach z_j: $z = (1-s)\, z_j$, $s \to 0^+$. Now, the dominant term satisfies:

$$F_\Sigma\big((1-s)\, z_j\big) \;=\; c_{j,1}\, s^{-\theta_{j,1}}\big[1 + o(1)\big]. \tag{4.21}$$

As a result,

$$\theta_{j,1} \;=\; \lim_{s\downarrow 0} \frac{\log\big|F_\Sigma\big((1-s)\, z_j\big)\big|}{-\log s}, \qquad c_{j,1} \;=\; \lim_{s\downarrow 0}\; s^{\theta_{j,1}}\, F_\Sigma\big((1-s)\, z_j\big). \tag{4.22}$$

Now, subtract the recovered term $c_{j,1}\Big[\big(1 - \frac{z}{z_j}\big)^{-\theta_{j,1}} - 1\Big]$ from F_Σ and repeat to recover the remaining exponents and coefficients at the same location. To recover the full set of atoms, repeat this at every singularity. ∎

Proposition 4.15 (Finite-data parameter counting)**.** *Given N complex observations, there are $2N$ real data values that constrain $5K$ real reconstruction parameters: τ_k, φ_k, θ_k, $\operatorname{Re} c_k$, $\operatorname{Im} c_k$. Therefore,*

$$2N \;\geq\; 5K. \tag{4.23}$$

This condition is necessary for unconstrained local identification.

Proof. Each of the complex coefficients a_{n_j} contributes two real values. Hence, the real data lies in $\mathbb{R}^{2N}$. Since each component has five parameters, the parameter space is of dimension $5K$. Therefore, for the coefficient map to be locally identifiable, its real Jacobian J_R must have full column rank $5K$, which is only possible if $5K \leq 2N$. ■

Definition 4.16 (Local identifiability condition)**.** The model is *locally identifiable* at a parameter point η if the Jacobian has full column rank there, after fixing the carrier-phase convention and the component ordering.

Remark: If positive gain and common time-origin shift are removed for the timbre-comparison (Definition 3.7), then the quotient has the dimension $5K - 2$. Still, actual reconstruction of the Fourier coefficients requires all $5K$ parameters.

4.4 Noise model and Fisher information

Definition 4.17 (Noise model)**.** Let ε denote independent circularly symmetric complex Gaussian noise:

$$y \;=\; A_N(\eta) + \varepsilon, \qquad \varepsilon \sim \mathcal{CN}\big(0, \sigma^2 I\big). \tag{4.24}$$

Here η is the real parameter vector, and A_N is the coefficient map of (3.2) evaluated at the corresponding signature.

Proposition 4.18 (Complex-Gaussian Fisher identity [11])**.** *For the (4.24) noise model, which has mean map $A_N(\eta)$ and complex Jacobian $J = \frac{\partial A_N}{\partial \eta}$, the finite-N Fisher information matrix is:*

$$I(\eta) \;=\; \frac{2}{\sigma^2}\,\mathrm{Re}\big(J^* J\big). \tag{4.25}$$

Proposition 4.19 (Nuisance elimination by Schur complement [11])**.** *We partition the parameter vector as $\eta = (\beta, \gamma)$, where γ contains the nuisance parameters and β the parameters of interest. The efficient information for β after eliminating γ is then the Schur complement:*

$$I_{eff} \;=\; I_{\beta\beta} \;-\; I_{\beta\gamma}\, I_{\gamma\gamma}^{\dagger}\, I_{\gamma\beta}. \tag{4.26}$$

5 The Dynamic Law: Time-Varying Signatures and Reconstruction

This section extends the fixed-time singularity signature of Sections 3 and 4 into a time-varying one. First, a windowed analysis convention is fixed to make the time-varying coefficients identifiable from a single waveform. Then, we establish the phase gauge, quantify the window bias and leakage, and finally prove the dynamic reconstruction theorem.

5.1 Windowed coefficient convention

The pointwise functions $a_n(t)$ cannot be uniquely recovered from a single scalar waveform unless the bandwidth of the signal is restricted or the analysis follows a fixed convention. Hence, we use the following windowed coefficients.

Definition 5.1 (Windowed heterodyne coefficient estimator)**.** Let w be a real, even, compactly supported function, normalised so that $\int w(s)\,ds = 1$. The analysis window w carries no component index, whereas the atom nodes w_k of (3.1) always carry the index k. Define

$$w_W(s) \;=\; W^{-1}\,w\Big(\frac{s}{W}\Big)\,. \tag{5.1}$$

With the independently fixed phase Φ, the *windowed heterodyne coefficient estimator* is

$$\tilde{a}_{n,W}(t) \;=\; \int_{\mathbb{R}} w_W(s)\,x(t+s)\,e^{-in\Phi(t+s)}\,ds. \tag{5.2}$$

Lemma 5.2 (Decomposition of the estimator)**.** *Let $x = x_H + x_R$ with x_H as in (2.7), and assume that the harmonic coefficients are absolutely summable. For every $n \geq 1$,*

$$\tilde{a}_{n,W}(t) \;=\; D_n(t) + \Lambda_n(t) + \Xi_n(t) + R_n(t), \tag{5.3}$$

where

$$D_n(t) \;=\; \int w_W(s)\,a_n(t+s)\,ds, \tag{5.4}$$

is the desired same-harmonic term,

$$\Lambda_n(t) = \sum_{m\geq 1,\, m\neq n} \int w_W(s)\,a_m(t+s)\,e^{i(m-n)\Phi(t+s)}\,ds, \tag{5.5}$$

is leakage from other positive harmonics,

$$\Xi_n(t) = \int w_W(s)\,a_0(t+s)\,e^{-in\Phi(t+s)}\,ds \;+\; \sum_{m\geq 1} \int w_W(s)\,\overline{a_m(t+s)}\,e^{-i(m+n)\Phi(t+s)}\,ds, \tag{5.6}$$

is conjugate-image leakage, and

$$R_n(t) \;=\; \int w_W(s)\,x_R(t+s)\,e^{-in\Phi(t+s)}\,ds, \tag{5.7}$$

is the residual-channel and noise term.

Proof. Substitute $x = x_H + x_R$ into (5.2) and use the two-sided harmonic expansion from Lemma 2.4. Absolute summability permits integration term-by-term, and the $m = n$ positive-frequency term contributes D_n, the $m \neq n$ terms Λ_n, the a_0 term and the conjugate-frequency terms contribute Ξ_n, and the residual term contributes R_n, which

together give (5.3). ■

5.2 Dynamic signature and phase gauge

Definition 5.3 (Dynamic signature)**.** A dynamic signature is a C^2 path

$$\Sigma\colon [0,T] \to S_K, \qquad \Sigma(t) = \Big\{ \big(\tau_k(t), \varphi_k(t), \theta_k(t), c_k(t)\big) \Big\}_{k=1}^{K}. \tag{5.8}$$

Combined with Φ, it produces

$$x_H(t) = a_0(t) + 2\,\mathrm{Re} \sum_{n\geq 1} \left[\sum_{k=1}^{K} c_k(t)\, b_n(\theta_k(t))\, e^{-\tau_k(t)\, n}\, e^{-in\varphi_k(t)} \right] e^{in\Phi(t)}. \tag{5.9}$$

Proposition 5.4 (Exact time-dependent gauge invariance)**.** *For any differentiable phase-shift function $\gamma\colon [0,T] \to \mathbb{R}$, the transformation*

$$\Phi(t) \mapsto \Phi(t) + \gamma(t), \qquad \varphi_k(t) \mapsto \varphi_k(t) + \gamma(t), \qquad k = 1, \dots, K, \tag{5.10}$$

leaves x_H unchanged.

Proof. For each n and k, $e^{-in\left(\varphi_k+\gamma\right)} e^{in(\Phi+\gamma)} = e^{-in\varphi_k} e^{in\Phi}$. Thus, every term in (5.9) remains unchanged. ■

5.3 Window bias and leakage

Theorem 5.5 (Symmetric-window $O\big(W^2\big)$ expansion)**.** *Let $a_n \in C^2$ near t. Let w be real, nonnegative, even, and supported in $\left[-\frac{1}{2}, \frac{1}{2}\right]$, with $\int w = 1$ and $\mu_2(w) = \int u^2 w(u)\, du$. Then*

$$D_n(t) - a_n(t) \;=\; \frac{W^2 \mu_2(w)}{2}\, \ddot{a}_n(t) \;+\; o\big(W^2\big), \qquad W \to 0. \tag{5.11}$$

Proof. The Taylor expansion gives $a_n(t+s) = a_n(t) + s\,\dot{a}_n(t) + \frac{s^2}{2}\,\ddot{a}_n(t) + s^2 \rho_n(t,s)$, $\rho_n(t,s) \to 0$. Using $s = Wu$ and the evenness of w, $\int w_W(s)\, ds = 1$, $\int s\, w_W(s)\, ds = 0$, and $\int s^2 w_W(s)\, ds = W^2 \mu_2(w)$. Let ε_W be the supremum of $\big|\rho_n(t,s)\big|$ over $|s| \leq \frac{W}{2}$. Since $\rho_n(t,s) \to 0$ as $s \to 0$, and w_W is supported in $\left[-\frac{W}{2}, \frac{W}{2}\right]$, we have $\varepsilon_W \to 0$ as $W \to 0$. Since $w \geq 0$, $\left|\int w_W(s)\, s^2 \rho_n(t,s)\, ds\right| \leq W^2 \mu_2(w)\, \varepsilon_W = o\big(W^2\big)$. Multiplying the Taylor expansion by $w_W(s)$ and integrating term by term, $D_n(t) = a_n(t) \cdot 1 + \dot{a}_n(t) \cdot 0 + \frac{\ddot{a}_n(t)}{2} \cdot W^2 \mu_2(w) + \int w_W(s)\, s^2 \rho_n(t,s)\, ds$, and subtracting $a_n(t)$ gives (5.11). ■

Proposition 5.6 (Leakage scale)**.** *Assume a locally constant carrier frequency $\Phi'(t) = \omega_0$. Define $\hat{w}(\omega) = \int w(u)\, e^{-i\omega u}\, du$ and $\hat{w}_W(\omega) = \hat{w}(W\omega)$. Then the leading leakage from harmonic $m \neq n$ is proportional to $\big|\hat{w}_W\big((m-n)\,\omega_0\big)\big|$, while the conjugate-image leakage is proportional to $\big|\hat{w}_W\big((m+n)\,\omega_0\big)\big|$.*

Proof. Since $\Phi(t+s) = \Phi(t) + \omega_0 s$, freezing $a_m(t+s)$ at $a_m(t)$ gives

$$\int w_W(s)\, a_m(t+s)\, e^{i(m-n)\Phi(t+s)}\, ds \approx a_m(t)\, e^{i(m-n)\Phi(t)}\, \hat{w}_W\big(-(m-n)\,\omega_0\big).$$

Since w is real and even, $\big|\hat{w}_W(-\omega)\big| = \big|\hat{w}_W(\omega)\big|$. The conjugate term is identical with $m-n$ replaced by $m+n$. ■

Proposition 5.7 (Coefficient-extraction error decomposition)**.** *For gradually varying carrier frequency, let $L_n(t,W) := \Lambda_n(t) + \Xi_n(t)$ and $\varepsilon_n(t) := R_n(t)$, with Λ_n, Ξ_n and R_n given in Lemma 5.2. By Lemma 5.2 and Theorem 5.5,*

$$\tilde{a}_{n,W}(t) - a_n(t) \;=\; \frac{W^2 \mu_2(w)}{2}\, \ddot{a}_n(t) \;+\; L_n(t,W) \;+\; \varepsilon_n(t) \;+\; o\big(W^2\big). \tag{5.12}$$

Proof. By Lemma 5.2, $\tilde{a}_{n,W}(t) = D_n(t) + \Lambda_n(t) + \Xi_n(t) + R_n(t)$. On subtracting $a_n(t)$ and grouping the leakage terms as $L_n(t,W)$ and the residual as $\varepsilon_n(t)$, we get $\tilde{a}_{n,W}(t) - a_n(t) = \big(D_n(t) - a_n(t)\big) + L_n(t,W) + \varepsilon_n(t)$. Now, substituting the expansion (5.11) of Theorem 5.5 for $D_n(t) - a_n(t)$ gives (5.12). ■

Hence, short windows increase leakage, while longer windows raise the temporal bias.

5.4 Quotient metric, joint objective, and trajectory lifting

Definition 5.8 (Quotient metric)**.** For $\Sigma, \Sigma' \in S_K$ with representatives $\big(h_k\big), \big(h'_k\big) \in H^K \setminus \Delta$, we define the *quotient metric*

$$d_Q\big(\Sigma, \Sigma'\big) \;=\; \min_{\pi \in \mathfrak{S}_K} \left(\sum_{k=1}^{K} \Big[\big|\tau_k - \tau'_{\pi(k)}\big|^2 + d_c\big(\varphi_k, \varphi'_{\pi(k)}\big)^2 + \big|\theta_k - \theta'_{\pi(k)}\big|^2 + \big|c_k - c'_{\pi(k)}\big|^2 \Big] \right)^{\frac{1}{2}}. \tag{5.13}$$

Here, $d_c(\alpha,\beta) = \min_{j\in\mathbb{Z}} \big|\alpha - \beta + 2\pi j\big|$ is the circular distance.

The value of d_Q is independent of the chosen representatives, and d_Q is a metric on S_K; both verifications are routine.

Definition 5.9 (Quotient-aware joint objective)**.** The *quotient-aware joint objective* is

$$J\big(\Sigma_1, \ldots, \Sigma_F\big) = J_0 + J_1 + J_2, \tag{5.14}$$

where:

$$J_0 \;=\; \sum_{j=1}^{F} \|\mathcal{W}_j^{\frac{1}{2}}\big(\tilde{a}_j - A_N\big(\Sigma_j\big)\big)\|_2^2, \tag{5.15}$$

$$J_1 = \lambda_1 \sum_{j=2}^{F} d_Q\big(\Sigma_j, \Sigma_{j-1}\big)^2, \qquad J_2 = \lambda_2 \sum_{j=2}^{F-1} \|\Delta_Q^2 \Sigma_j\|_G^2. \tag{5.16}$$

Here, d_Q is the quotient metric of Definition 5.8, $\|\cdot\|_G$ denotes the Euclidean norm on labelled parameter vectors, and $\Delta_Q^2\Sigma_j$ is the labelled parameter vector defined after a locally consistent lift of the individual frames.

Proposition 5.10 (Path lifting away from collisions)**.** *Let Δ denote the collision set. The action of $\mathfrak{S}_K$ on $H^K \setminus \Delta$ is free. Hence, the quotient map $pr\colon H^K \setminus \Delta \to S_K$ is a $K!$-sheeted covering map, and every continuous path $\Sigma\colon [0,T] \to S_K$ has a unique continuous lift $\tilde{\Sigma}\colon [0,T] \to H^K \setminus \Delta$ once $\tilde{\Sigma}(0) \in pr^{-1}\big(\Sigma(0)\big)$ is chosen.*

Proof. Immediate from the theory of covering spaces; see [8]. ■

At collisions, the labelling of the trajectories is only defined up to permutation. Hence, labelled trajectories are only considered on intervals that do not contain collisions.

5.5 The dynamic reconstruction theorem

Theorem 5.11 (Dynamic reconstruction with explicit error chain)**.** *Fix N and a carrier phase Φ. Let the true C^2 trajectory lie in a compact, gauge-fixed, and collision-free chart K, and suppose $\sigma_{min}\big(J_R(\Sigma)\big) \geq \sigma_0 > 0$ on K. Suppose further that a local least-squares solution to the dynamic inverse problem is found in the normal neighbourhood of the true trajectory, and that the frame residuals satisfy*

$$\|\tilde{a}_W(t_j) - A_N\big(\Sigma(t_j)\big)\|_2 \;\leq\; \rho_W, \tag{5.17}$$

with ρ_W given by

$$\rho_W \;=\; C_b W^2 M_2 \;+\; C_L(W) \;+\; \delta \;+\; o\big(W^2\big). \tag{5.18}$$

Here, M_2 denotes the maximum of $\|\ddot{a}(t)\|_2$ on $[0,T]$, $C_b = \frac{\mu_2(w)}{2}$ is the within-window bias constant of Theorem 5.5, $C_L(W) = \sup_{n,t}\big|L_n(t,W)\big|$ is the leakage magnitude of Proposition 5.7, and δ is the noise level.

Then, for sufficiently small ρ_W,

$$d_Q\big(\hat{\Sigma}(t_j), \Sigma(t_j)\big) \;\leq\; \frac{4\rho_W}{\sigma_0}. \tag{5.19}$$

If the trajectory is interpolated using a second-order accurate quotient-consistent interpolant with hop h, the error in the coefficient path is at most

$$L_A\Big(\frac{4\rho_W}{\sigma_0} + Ch^2\Big), \tag{5.20}$$

where L_A is a local Lipschitz constant of A_N.

Proof. By Theorem 5.5 and Proposition 5.7, the true signature has residual no larger than ρ_W. Thus, the local least-squares minimiser has a residual no larger, hence: $\|A_N\big(\hat{\Sigma}(t_j)\big) -$

$A_N\big(\Sigma(t_j)\big)\|_2 \le 2\rho_W$. In the labelled coordinate neighbourhood provided by Proposition 5.10, let η_1, η_2 lie in a sufficiently small convex neighbourhood. Then, $A_N\big(\eta_1\big) - A_N\big(\eta_2\big) = \Big[\int_0^1 J_R\big(\eta_2 + s\big(\eta_1 - \eta_2\big)\big)\, ds\Big]\big(\eta_1 - \eta_2\big)$. By continuity of J_R, the neighbourhood may be chosen such that the averaged Jacobian value has the smallest singular value at least $\frac{\sigma_0}{2}$. Thus $\|A_N\big(\eta_1\big) - A_N\big(\eta_2\big)\|_2 \ge \frac{\sigma_0}{2}\|\eta_1 - \eta_2\|_2$. Since d_Q is the minimum over $\pi \in \mathfrak{S}_K$ of the labelled distance, it is at most the labelled distance for the identity permutation. Combining the preceding bounds gives $d_Q\big(\hat{\Sigma}(t_j), \Sigma(t_j)\big) \le \|\hat{\eta}(t_j) - \eta(t_j)\|_2 \le \frac{4\rho_W}{\sigma_0}$, establishing (5.19). The second-order interpolant introduces an error of order Ch^2 in parameter space, and the local Lipschitz condition for A_N establishes the stated coefficient-path estimate. ■

Remark: Near collisions or vanishing components, σ_0 may approach zero and parameter error could be high even if the coefficients are accurately reconstructed.

5.6 Truncation and energy bounds

Lemma 5.12 (Parseval identity on the frozen cycle [10])**.** *With normalised measure* $\frac{du}{2\pi}$,

$$\|g_t - \hat{g}_{t,N}\|^2_{L^2(\mathbb{T})} \;=\; 2\sum_{n=1}^{N}\big|a_n(t) - \hat{a}_n(t)\big|^2 \;+\; 2\sum_{n>N}\big|a_n(t)\big|^2, \tag{5.21}$$

where $\hat{g}_{t,N}(u) = \hat{a}_0(t) + 2\,\mathrm{Re}\sum_{n=1}^{N}\hat{a}_n(t)\,e^{inu}$ *denotes the* N*-term truncation of the fitted frozen-cycle waveform, and* a_0 *is assumed exact or treated separately.*

The first term is fitted-band error; the second term is harmonic truncation.

Proposition 5.13 (Truncation-tail bound)**.** *Assume that* $\tau_k \ge \tau_{min} > 0$, $\theta_k \le \theta_{max}$, *and that* $\big|c_k\big|$ *is uniformly bounded on a compact set of trajectories. Then*

$$\sum_{n>N}\big|a_n(t)\big|^2 \;=\; O\big(N^{2\theta_{max}-2}\,e^{-2\tau_{min}N}\big) \tag{5.22}$$

uniformly on that set.

Proof. By the triangle inequality, $\big|a_n(t)\big| \le \sum_k \big|c_k\big|\, b_n(\theta_k)\, e^{-\tau_k n}$. Since $e^{-\tau n}$ is nonincreasing in τ and $n^{\theta-1}$ is nondecreasing in θ for $n \ge 1$, each of the K summands is bounded by $Cn^{\theta_{max}-1}e^{-\tau_{min}n}$ by Proposition 3.5 with C uniform on the compact set of trajectories. Summing the K terms gives $\big|a_n(t)\big| \le Cn^{\theta_{max}-1}e^{-\tau_{min}n}$. Setting $s_0 = 2\theta_{max} - 2$ and $c_0 = 2\tau_{min}$, it follows that $\sum_{n>N}\big|a_n(t)\big|^2 \le C^2\sum_{n>N} n^{s_0}e^{-c_0 n}$. For sufficiently large N, $n^{s_0}e^{-\frac{c_0 n}{2}}$ is decreasing, so $\sum_{n>N} n^{s_0}e^{-c_0 n} \le \frac{(N+1)^{s_0}\,e^{-c_0(N+1)}}{1-e^{-\frac{c_0}{2}}} = O\big(N^{s_0}e^{-c_0 N}\big)$. Substituting the values of s_0 and c_0 gives (5.22). ■

Time-domain reconstruction additionally depends on carrier-phase accuracy and temporal interpolation, with the associated errors treated separately.

6 Timbre-Descriptor Calculus

6.1 Brightness: the magnitude centroid and its monotonicity

Definition 6.1 (Component weights and centroid)**.** For a single component, define the nonnegative magnitude-envelope weights

$$\kappa_n \;=\; b_n(\theta)\, y^n, \qquad y = e^{-\tau}, \qquad n \geq 1. \tag{6.1}$$

Its centroid, measured in harmonic units, is

$$C_1 \;=\; \frac{\sum_{n\geq 1} n\,\kappa_n}{\sum_{n\geq 1} \kappa_n}. \tag{6.2}$$

Theorem 6.2 (Closed form of the isolated centroid)**.** *For $\tau > 0$ and $\theta > 0$, let $y = e^{-\tau} \in (0,1)$. Then*

$$C_1(\tau,\theta) \;=\; \frac{\theta y(1-y)^{-\theta-1}}{(1-y)^{-\theta}-1} \;=\; \frac{\theta y}{(1-y)\big[1-(1-y)^{\theta}\big]}. \tag{6.3}$$

Proof. For the denominator, Theorem 2.10 and $b_0(\theta) = 1$ from Lemma 2.9(i) give

$$\sum_{n=1}^{\infty} b_n(\theta)\, y^n = \sum_{n=0}^{\infty} b_n(\theta)\, y^n - 1 = (1-y)^{-\theta} - 1.$$

For the numerator, the power series can be differentiated term by term inside its disk of convergence. Series (2.13) has radius 1. Differentiating the $n \geq 0$ generating series with respect to y and multiplying by y gives

$$y\frac{d}{dy}(1-y)^{-\theta} = y\theta(1-y)^{-\theta-1} = \sum_{n=0}^{\infty} n\, b_n(\theta)\, y^n = \sum_{n=1}^{\infty} n\, b_n(\theta)\, y^n.$$

Here $\frac{d}{dy}(1-y)^{-\theta} = \theta(1-y)^{-\theta-1}$ by the chain rule, and the $n = 0$ term vanishes because it is multiplied by n. Therefore $C_1(\tau,\theta) = \frac{\theta y(1-y)^{-\theta-1}}{(1-y)^{-\theta}-1}$. To obtain the second form in (6.3), factor the denominator as $(1-y)^{-\theta} - 1 = (1-y)^{-\theta}\big[1-(1-y)^{\theta}\big]$ and cancel:

$$C_1 = \frac{\theta y(1-y)^{-\theta-1}}{(1-y)^{-\theta}\big[1-(1-y)^{\theta}\big]} = \frac{\theta y(1-y)^{-1}}{1-(1-y)^{\theta}} = \frac{\theta y}{(1-y)\big[1-(1-y)^{\theta}\big]}.$$

■

Proposition 6.3 (Uniform shallow-depth asymptotic)**.** *Because $1-e^{-\tau} = \tau + O\big(\tau^2\big)$, the leading-order statement, uniform in every $\theta > 0$, is*

$$C_1(\tau,\theta) \;=\; \frac{\theta}{\tau}\Big[1 + O\big(\tau^{min(1,\theta)}\big)\Big] \qquad \text{as } \tau \downarrow 0. \tag{6.4}$$

Proof. Put $y = e^{-\tau}$ into the second expression in (6.3). Expanding the exponential gives $1-y = \tau - \frac{\tau^2}{2} + O\big(\tau^3\big) = \tau\big(1+O(\tau)\big)$ and $y = 1+O(\tau)$. Also, $(1-y)^\theta = \tau^\theta\big(1+O(\tau)\big)^\theta = \tau^\theta\big(1+O(\tau)\big)$, so $\frac{1}{1-(1-y)^\theta} = \frac{1}{1-\tau^\theta(1+O(\tau))} = 1+O\big(\tau^\theta\big)$. Combining the three factors gives $C_1 = \frac{\theta(1+O(\tau))}{\tau(1+O(\tau))}\big(1+O\big(\tau^\theta\big)\big) = \frac{\theta}{\tau}(1+O(\tau))\big(1+O\big(\tau^\theta\big)\big) = \frac{\theta}{\tau}\Big[1+O\big(\tau^{min(1,\theta)}\big)\Big]$, because the larger of the relative corrections $O(\tau)$ and $O\big(\tau^\theta\big)$ controls the result. ■

Remark: The commonly written additive remainder $C_1 = \frac{\theta}{\tau} + O(1)$ is not valid when $0 < \theta < 1$. From (6.4), the next absolute term is $\frac{\theta}{\tau}\cdot O\big(\tau^\theta\big) = O\big(\tau^{\theta-1}\big)$, which diverges as $\tau \downarrow 0$ for $\theta < 1$. Hence, the relative approximation $C_1 \approx \frac{\theta}{\tau}$ is still valid.

Normalising the weights, $p_n = \frac{\kappa_n}{\sum_{m\geq 1}\kappa_m}$ defines a probability distribution on $n \geq 1$. Write E_p, Var_p, and Cov_p for its moments. Then $C_1 = E_p[n]$.

Theorem 6.4 (Exact depth monotonicity)**.** *For every $\tau > 0$ and $\theta > 0$,*

$$\partial_\tau C_1 \;=\; -\,\mathrm{Var}_p(n) \;<\; 0. \tag{6.5}$$

Proof. From (6.1), $\partial_\tau \kappa_n = \partial_\tau\big[b_n(\theta)\, e^{-\tau n}\big] = -\,n\,\kappa_n$. Applying the quotient rule to $p_n = \frac{\kappa_n}{\sum_m \kappa_m}$ gives $\partial_\tau p_n = \frac{(-n\kappa_n)\sum_m \kappa_m - \kappa_n \sum_m(-m\kappa_m)}{\big(\sum_m \kappa_m\big)^2} = p_n\big(E_p[n] - n\big)$. Therefore $\partial_\tau C_1 = \sum_n n\,\partial_\tau p_n = \sum_n n\,p_n\big(E_p[n]-n\big) = E_p[n]^2 - E_p\big[n^2\big] = -\,\mathrm{Var}_p(n)$. The distribution is nondegenerate because it assigns positive weight to every $n \geq 1$, so its variance is strictly positive. ■

Theorem 6.5 (Exact type monotonicity)**.** *Define the θ-score $s_n(\theta) = \psi(n+\theta) - \psi(\theta)$. Then*

$$\partial_\theta C_1 \;=\; \mathrm{Cov}_p\big(n, s_n\big) \;>\; 0. \tag{6.6}$$

Proof. Proposition 4.12 gives $\partial_\theta \kappa_n = \kappa_n\, s_n(\theta)$. Repeating the quotient-rule calculation from Theorem 6.4, with $-n$ replaced by s_n, gives $\partial_\theta p_n = p_n\big(s_n - E_p[s]\big)$. Hence $\partial_\theta C_1 = \sum_n n\,p_n\big(s_n - E_p[s]\big) = E_p\big[n s_n\big] - E_p[n]\,E_p\big[s_n\big] = \mathrm{Cov}_p\big(n, s_n\big)$. To prove positivity, use the digamma recurrence $\psi(z+1) = \psi(z) + \frac{1}{z}$ [14]: $s_{n+1} - s_n = \psi(n+1+\theta) - \psi(n+\theta) = \frac{1}{n+\theta} > 0$. Both n and s_n therefore increase strictly with n. Their covariance under a nondegenerate distribution is strictly positive: using independent $n_1, n_2 \sim p$ and expanding the product, the exchange identity is $\mathrm{Cov}_p\big(n, s_n\big) = \frac{1}{2}E\Big[\big(n_1 - n_2\big)\big(s_{n_1} - s_{n_2}\big)\Big] = \frac{1}{2}\sum_{i,j} p_i p_j (i-j)\big(s_i - s_j\big)$. Every term is nonnegative because $(i-j)$ and $\big(s_i - s_j\big)$ have the same sign, and the terms with $i \neq j$ are strictly positive. Thus $\partial_\theta C_1 > 0$. ■

Corollary 6.6 (Brightness dynamics along a path)**.** *Along a C^1 path $\big(\tau(t), \theta(t)\big)$,*

$$\dot{C}_1 \;=\; \mathrm{Cov}_p\big(n, s_n\big)\,\dot{\theta} \;-\; \mathrm{Var}_p(n)\,\dot{\tau}. \tag{6.7}$$

Thus $\dot{\tau} < 0$ yields strict brightening when $\dot{\theta} \geq 0$. If θ also decreases, brightening occurs exactly when $\mathrm{Var}_p(n)\big(-\dot{\tau}\big) > \mathrm{Cov}_p\big(n, s_n\big)\big(-\dot{\theta}\big)$.

Proof. The chain rule gives $\dot{C}_1 = \big(\partial_\theta C_1\big)\dot{\theta} + \big(\partial_\tau C_1\big)\dot{\tau}$. Substituting (6.5) and (6.6) yields (6.7). When $\dot{\theta} \geq 0$ and $\dot{\tau} < 0$, both terms are nonnegative and the second is positive. In the mixed case, $\dot{C}_1 > 0$ exactly when the variance term is larger than the covariance term. ■

Remark: Differentiate the approximation $C_1 \approx \frac{\theta}{\tau}$ from Proposition 6.3 with respect to t. The quotient rule gives

$$\dot{C}_1 \;\approx\; \frac{\dot{\theta}}{\tau} \;-\; \frac{\theta\,\dot{\tau}}{\tau^2}. \tag{6.8}$$

6.2 The energy centroid of the full mixture

Classical spectral centroids often use power weights. In the complete mixture, phase cancellation means that isolated-component centroids cannot simply be added. An exact formula for the mixture is still available.

Definition 6.7 (Energy and first moment)**.** For the full signature, set

$$E(\Sigma) \;=\; \sum_{n\geq 1} \big|a_n\big|^2, \qquad M_1(\Sigma) \;=\; \sum_{n\geq 1} n\big|a_n\big|^2, \qquad C_2(\Sigma) \;=\; \frac{M_1(\Sigma)}{E(\Sigma)}. \tag{6.9}$$

The energy-weighted centroid is $C_2(\Sigma)$.

Proposition 6.8 (Gauss hypergeometric function [14])**.** *For parameters a and b with $|\zeta| < 1$,*

$$ {}_2F_1\big(a, b; 1; \zeta\big) \;=\; \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(n!)^2}\, \zeta^n, \tag{6.10}$$

and the contiguous derivative identity is

$$\frac{d}{d\zeta}\, {}_2F_1\big(a, b; c; \zeta\big) \;=\; \frac{ab}{c}\, {}_2F_1\big(a+1, b+1; c+1; \zeta\big). \tag{6.11}$$

Both identities are standard and are used without proof.

Theorem 6.9 (Exact mixture energy)**.** *Write $a_n = \sum_k c_k\, b_n(\theta_k)\, w_k^n$ and $\zeta_{kl} = w_k \overline{w_l}$, so $\big|\zeta_{kl}\big| = e^{-(\tau_k + \tau_l)} < 1$. Then*

$$E(\Sigma) \;=\; \sum_{k,l} c_k \overline{c_l} \Big[{}_2F_1\big(\theta_k, \theta_l; 1; \zeta_{kl}\big) - 1 \Big]. \tag{6.12}$$

Proof. Expand the energy using $|z|^2 = z\bar{z}$ and the mixture law:

$$\begin{aligned}\sum_{n=1}^{\infty}\big|a_n\big|^2 &= \sum_{n=1}^{\infty}\Big(\sum_{k} c_k\, b_n(\theta_k)\, w_k^{\,n}\Big)\Big(\sum_{l} \overline{c_l}\, b_n(\theta_l)\, \overline{w_l}^{\,n}\Big)\\ &= \sum_{n=1}^{\infty}\sum_{k,l} c_k\overline{c_l}\, b_n(\theta_k)\, b_n(\theta_l)\, \zeta_{kl}^{\,n}.\end{aligned}$$

Here $w_k^{\,n}\overline{w_l}^{\,n} = \big(w_k\overline{w_l}\big)^n = \zeta_{kl}^{\,n}$. The double series converges absolutely: $\big|\zeta_{kl}\big| \le e^{-2\tau_{min}} < 1$ and $b_n(\theta) = O\big(n^{\theta-1}\big)$ by Proposition 2.11. Each inner series is therefore dominated by a convergent geometric-polynomial series, while the outer sum is finite. The order of summation may be exchanged: $E(\Sigma) = \sum_{k,l} c_k\overline{c_l}\sum_{n=1}^{\infty}\frac{(\theta_k)_n(\theta_l)_n}{(n!)^2}\,\zeta_{kl}^{\,n}$. Lemma 2.9(ii) gives $b_n(\theta_k)\, b_n(\theta_l) = \frac{(\theta_k)_n(\theta_l)_n}{(n!)^2}$. By (6.10), the inner sum from $n=1$ equals ${}_2F_1\big(\theta_k,\theta_l;1;\zeta_{kl}\big) - 1$, since the $n=0$ term is 1. This proves (6.12). ∎

Theorem 6.10 (Exact mixture first moment and centroid)**.** *Using the notation of Theorem 6.9,*

$$M_1(\Sigma) \;=\; \sum_{k,l} c_k\overline{c_l}\,\theta_k\theta_l\,\zeta_{kl}\;{}_2F_1\big(\theta_k+1,\theta_l+1;2;\zeta_{kl}\big), \tag{6.13}$$

and the full energy-weighted centroid is

$$C_2(\Sigma) \;=\; \frac{M_1(\Sigma)}{E(\Sigma)}. \tag{6.14}$$

Proof. Apply $\zeta\frac{d}{d\zeta}$ to (6.10). Term-by-term differentiation inside the disk gives

$$\begin{aligned}\zeta\frac{d}{d\zeta}\,{}_2F_1\big(a,b;1;\zeta\big) &= \sum_{n=0}^{\infty} n\,\frac{(a)_n(b)_n}{(n!)^2}\,\zeta^n\\ &= \sum_{n=1}^{\infty} n\,\frac{(a)_n(b)_n}{(n!)^2}\,\zeta^n.\end{aligned}$$

This is the weighted sum required for the first moment. By (6.11) with $c=1$,

$$\zeta\frac{d}{d\zeta}\,{}_2F_1\big(a,b;1;\zeta\big) = \zeta\,ab\;{}_2F_1\big(a+1,b+1;2;\zeta\big).$$

Repeating the expansion in Theorem 6.9 with the extra factor n gives

$$\begin{aligned}M_1(\Sigma) &= \sum_{k,l} c_k\overline{c_l}\sum_{n\ge1} n\,\frac{(\theta_k)_n(\theta_l)_n}{(n!)^2}\,\zeta_{kl}^{\,n}\\ &= \sum_{k,l} c_k\overline{c_l}\,\zeta_{kl}\,\theta_k\theta_l\;{}_2F_1\big(\theta_k+1,\theta_l+1;2;\zeta_{kl}\big).\end{aligned}$$

The same absolute-convergence argument allows the sum to be reordered; the extra factor

n remains summable against the geometric factor. Dividing by (6.12) gives (6.14). ■

Proposition 6.11 (Realness)**.** *Individual cross terms may be complex, but conjugate symmetry makes E and M_1 real.*

Proof. Compare the (k,l) term of (6.12) with the (l,k) term. Since $\zeta_{lk} = w_l\overline{w_k} = \overline{\zeta_{kl}}$, and the ${}_2F_1$ series (6.10) has real coefficients for real θ_k and θ_l, ${}_2F_1\big(\theta_l,\theta_k;1;\zeta_{lk}\big)$ is the conjugate of ${}_2F_1\big(\theta_k,\theta_l;1;\zeta_{kl}\big)$; symmetry in the first two arguments is also used. Likewise, $c_l\overline{c_k}$ is the conjugate of $c_k\overline{c_l}$. Thus the (l,k) term is the complex conjugate of the (k,l) term. Diagonal terms $k = l$ are real, so the sum equals its own conjugate. The same argument applies to (6.13). ■

The formula includes all interference between components. It is therefore the correct exact descriptor when a centroid is claimed for the complete signature rather than for one isolated atom.

6.3 Spectral shape and envelope: tilt, curvature, and peak

Proposition 6.12 (Log-envelope and asymptotic slope)**.** *For one component, taking logarithms of the asymptotic envelope (3.7) gives*

$$\log\big|a_n\big| \;=\; \log|c| - \log\Gamma(\theta) + (\theta-1)\log n - \tau n + O\big(n^{-1}\big), \tag{6.15}$$

and, if n is treated as continuous when taking the asymptotic slope,

$$\frac{d}{dn}\log\big|a_n\big| \;=\; -\tau \;+\; \frac{\theta-1}{n} \;+\; O\big(n^{-2}\big). \tag{6.16}$$

Proof. In (3.7), the product of $\frac{|c|}{\Gamma(\theta)}$, $n^{\theta-1}$, $e^{-\tau n}$, and $\big[1+O\big(n^{-1}\big)\big]$ becomes a sum after logarithms are taken. The Mercator expansion gives $\log\big[1+O\big(n^{-1}\big)\big] = O\big(n^{-1}\big)$, which is (6.15). For (6.16) we differentiate the exact expression rather than the remainder, since an $O\big(n^{-1}\big)$ term need not have an $O\big(n^{-2}\big)$ derivative. By Definition 2.8, $\log\big|a_n\big| = \log|c| + \log\Gamma(n+\theta) - \log\Gamma(\theta) - \log\Gamma(n+1) - \tau n$, so treating n as continuous gives exactly

$$\frac{d}{dn}\log\big|a_n\big| \;=\; \psi(n+\theta) - \psi(n+1) - \tau.$$

The digamma expansion $\psi(x) = \log x - \frac{1}{2x} + O\big(x^{-2}\big)$ [14] then gives $\psi(n+\theta) - \psi(n+1) = \log\frac{n+\theta}{n+1} + O\big(n^{-2}\big) = \frac{\theta-1}{n} + O\big(n^{-2}\big)$, which is (6.16). ■

Thus τ gives the high-harmonic log-linear tilt, while θ controls the algebraic curvature. For a multi-component mixture, destructive interference and changes in the dominant component can make one global slope invalid. In that case, compute the descriptor from the generated coefficients or report values for each component.

Proposition 6.13 (Peak location of the isolated envelope)**.** *Let* $\kappa_n = b_n(\theta)\, e^{-\tau n}$*. Then*

$$\frac{\kappa_{n+1}}{\kappa_n} \;=\; e^{-\tau}\,\frac{n+\theta}{n+1}, \tag{6.17}$$

and the sequence increases from n *to* $n+1$ *exactly when*

$$n \;\le\; \nu := \frac{\theta - e^{\tau}}{e^{\tau}-1}. \tag{6.18}$$

Therefore, for $n \ge 1$*, one valid peak index is*

$$n^{*} \;=\; \max\Big\{1,\; \lceil \tfrac{\theta-e^{\tau}}{e^{\tau}-1} \rceil \Big\}. \tag{6.19}$$

Proof. The ratio recurrence (2.12) gives $\frac{\kappa_{n+1}}{\kappa_n} = \left[\frac{b_{n+1}(\theta)}{b_n(\theta)}\right] \frac{e^{-\tau(n+1)}}{e^{-\tau n}} = e^{-\tau}\,\frac{n+\theta}{n+1}$, proving (6.17). The step is nondecreasing precisely when the ratio is at least one: $e^{-\tau}(n+\theta) \ge n+1 \Leftrightarrow n+\theta \ge e^{\tau}(n+1) \Leftrightarrow n\big(1-e^{\tau}\big) \ge e^{\tau}-\theta \Leftrightarrow n \le \frac{e^{\tau}-\theta}{1-e^{\tau}} = \frac{\theta-e^{\tau}}{e^{\tau}-1}$. The inequality reverses because $1-e^{\tau} < 0$ when $\tau > 0$; the final equality multiplies numerator and denominator by -1. The sequence increases while $n \le \nu$ and then decreases, so the first integer above ν is a peak. Clipping at 1 accounts for the measured harmonic range beginning at $n = 1$. ∎

Remark: If ν is a positive integer, $\kappa_{\nu} = \kappa_{\nu+1}$, so both adjacent indices are maxima. Formula (6.19) selects the lower one. When $\tau \to 0$ and $\theta > 1$, n^{*} is asymptotic to $\frac{\theta-1}{\tau}$. Expanding $e^{\tau} = 1 + \tau + O\big(\tau^{2}\big)$ in (6.18) gives $\nu = \frac{\theta-1-\tau+O(\tau^{2})}{\tau+O(\tau^{2})} \to \frac{\theta-1}{\tau}$. The $\max\{1,\cdot\}$ correction is needed whenever the formal ν lies below the measured harmonic range.

6.4 Temporal descriptors: attack, decay, modulation, and spectral flux

Theorem 6.14 (Log-magnitude rate of a component coefficient)**.** *For*

$$q_{n,k} = c_k\, b_n(\theta_k)\, e^{-\tau_k n}\, e^{-in\varphi_k}$$

along a C^{1} *parameter path,*

$$\begin{aligned} \frac{d}{dt}\log\big|q_{n,k}\big| \;&=\; \mathrm{Re}\Big(\frac{\dot c_k}{c_k}\Big) \\ &\quad + \big[\psi\big(n+\theta_k\big) - \psi\big(\theta_k\big)\big]\dot\theta_k - n\,\dot\tau_k. \end{aligned} \tag{6.20}$$

Proof. Take the logarithm of the modulus. Since $\big|e^{-in\varphi_k}\big| = 1$, the angle term disappears: $\log\big|q_{n,k}\big| \;=\; \log\big|c_k\big| + \log b_n(\theta_k) - \tau_k n$. Differentiate the three terms with respect to t. For amplitude, write $\log|c| = \frac{1}{2}\log(c\bar c)$; then $\frac{d}{dt}\log|c| = \frac{1}{2}\Big(\frac{\dot c}{c} + \frac{\dot{\bar c}}{\bar c}\Big) = \mathrm{Re}\Big(\frac{\dot c}{c}\Big)$. For type, Proposition 4.12 and the chain rule give $\frac{d}{dt}\log b_n(\theta_k(t)) = \big[\psi\big(n+\theta_k\big) - \psi\big(\theta_k\big)\big]\dot\theta_k$. For

depth, $\frac{d}{dt}\big(-\tau_k n\big) = -\,n\,\dot\tau_k$. Adding the three contributions gives (6.20). ∎

This separates changes in gain, type or curvature, and depth. If $\dot\tau_k < 0$, then $-n\dot\tau_k$ is positive and grows linearly with harmonic number, so the component brightens. If $\big|c_k\big|$ falls while τ_k rises, both overall strength and high-harmonic content decrease, but through different mechanisms. Any statement about $\big|c_k\big|$ alone remains conditional on τ_k and θ_k because all three affect total component energy.

Proposition 6.15 (Instantaneous phase and frequency of a component harmonic)**.** *Let $c_k = \big|c_k\big|\exp\big(i\chi_k\big)$. A single component contributes the phase*

$$\Psi_{n,k}(t) \;=\; n\Phi(t) - n\varphi_k(t) + \chi_k(t) \tag{6.21}$$

to harmonic n. Wherever $c_k \neq 0$, its instantaneous angular frequency is exactly

$$\Omega_{n,k} \;=\; n\Phi'(t) - n\dot\varphi_k(t) + \dot\chi_k(t), \tag{6.22}$$

or, in hertz,

$$f_{n,k} \;=\; nf_0(t) - \frac{n\dot\varphi_k(t)}{2\pi} + \frac{\dot\chi_k(t)}{2\pi}. \tag{6.23}$$

Proof. Component k contributes $q_{n,k}(t)\,e^{in\Phi(t)}$ to harmonic n of the analytic representation. The arguments of its factors are $\arg\big(c_k\big) = \chi_k$, $\arg\Big[b_n(\theta_k)\,e^{-\tau_k n}\Big] = 0$ because that factor is positive and real, $\arg\big(e^{-in\varphi_k}\big) = -\,n\varphi_k$, and $\arg\big(e^{in\Phi}\big) = n\Phi$. Their sum gives (6.21). Instantaneous angular frequency is the time derivative of unwrapped phase, so differentiating (6.21) gives (6.22). This is well defined wherever $c_k \neq 0$, where χ_k is locally smooth. Dividing by 2π and using $\Phi' = 2\pi f_0$ gives (6.23). ∎

The φ term scales with n and represents coherent modulation once the Φ gauge has been fixed. The phase of c is shared across n for that component. If components interfere, the instantaneous phase of their sum is not the sum of their separate phase laws.

Theorem 6.16 (Pullback-metric identity)**.** *Let $\eta(t)$ be a real local coordinate vector on a labelled chart and let $a(t) = A_N(\eta(t))$. Then*

$$\dot a \;=\; J(\eta(t))\,\dot\eta(t), \tag{6.24}$$

and, for nonnegative harmonic weights $\mathcal{W}$,

$$\|\mathcal{W}^{\frac12}\dot a\|_2^2 \;=\; \dot\eta^{T}\,\mathrm{Re}\big(J^{*}\mathcal{W}J\big)\,\dot\eta. \tag{6.25}$$

Proof. Equation (6.24) is the chain rule applied component by component to $a_j(t) = A_{n_j}(\eta(t))$: $\dot a_j = \sum_i \big(\frac{\partial A_{n_j}}{\partial \eta_i}\big)\dot\eta_i$. In matrix form, this is $J\dot\eta$, where J is the complex Jacobian in Definition 4.13. For (6.25), substitute (6.24) and expand the weighted norm using

$\|v\|_2^2 = v^*v$: $\|\mathcal{W}^{\frac{1}{2}}\dot{a}\|_2^2 = \dot{a}^*\mathcal{W}\dot{a} = \big(J\dot{\eta}\big)^*\mathcal{W}\big(J\dot{\eta}\big) = \dot{\eta}^T\big(J^*\mathcal{W}J\big)\dot{\eta}$, because $\dot{\eta}$ is real. The matrix $M = J^*\mathcal{W}J$ is Hermitian. Write $M = \operatorname{Re} M + i \operatorname{Im} M$. Hermitian symmetry makes $\operatorname{Re} M$ symmetric and $\operatorname{Im} M$ antisymmetric. For a real vector v and antisymmetric A, v^TAv equals its transpose $v^TA^Tv = -v^TAv$, so it is zero. Therefore $\dot{\eta}^TM\dot{\eta} = \dot{\eta}^T(\operatorname{Re} M)\dot{\eta} + i\,\dot{\eta}^T(\operatorname{Im} M)\dot{\eta} = \dot{\eta}^T(\operatorname{Re} M)\dot{\eta} = \dot{\eta}^T \operatorname{Re}\big(J^*\mathcal{W}J\big)\dot{\eta}$, which proves (6.25) and is real, as a squared norm must be. ■

The matrix $\operatorname{Re}\big(J^*\mathcal{W}J\big)$ therefore has three roles: it is the pullback metric for spectrotemporal motion, the local inverse-conditioning matrix, and, under circular Gaussian noise, Fisher information up to the factor $\frac{2}{\sigma^2}$ from Proposition 4.18. The identity connects descriptor motion with statistical identifiability. Parameter directions that change the spectrum only slightly have low flux and are difficult to estimate.

7 Applications of the Model

7.1 Analysis and resynthesis

The principal generative test is the closed chain

$$audio \;\to\; \big(\hat{\Phi},\; \hat{a}_n\big(t_j\big)\big) \;\to\; \hat{\Sigma}\big(t_j\big) \;\to\; \hat{a}_n(t) \;\to\; \hat{x}_H(t). \tag{7.1}$$

The protocol has four steps.

1. At the source sample rate, downmix the recording to a single channel. Do this by averaging the stereo channels.
2. Using the YIN cumulative-mean-normalised difference method [2], estimate the constant fundamental frequency for the whole note f_0.
3. Extract 17 Hann-windowed heterodyne frames. Each window is capped at 120 ms and is the longer of 25 ms and four fundamental periods.
4. Interpolate the coefficients within the frames and resynthesise the signal.

The extracted coefficients represent the analysis component of the signal. This component excludes noise and inharmonic energy, but it retains the window leakage and coefficient-estimation noise.

7.2 Trajectory storage and rate sensitivity

Proposition 7.1 (State counts)**.** *An unconstrained N-harmonic frame contains $2N$ real coefficient values. A K-component signature contains*

$$K \times \big(\tau,\; \varphi,\; \theta,\; \operatorname{Re} c,\; \operatorname{Im} c\big) \;=\; 5K \tag{7.2}$$

real values. For F independently fitted frames, the nominal state count is $5KF$, excluding the carrier, time stamps, quantisation, and residual channel. Furthermore, if each of the $5K$ parameter curves is represented by its values at Q prescribed temporal samples and reconstructed by linear interpolation between adjacent samples, the nominal trajectory count is $5KQ$ real values, excluding the shared sample times and metadata.

Proof. Each complex coefficient comprises two real numbers, so a frame of N coefficients contributes $2N$. Each component of a signature contains the five real coordinates: depth, angle, type, and the two real coordinates of c, resulting in $5K$ parameters per signature. Multiplying one frame's parameter count by the number F of independent frames gives the state count $5KF$. Representing each of the $5K$ parameter curves by Q retained temporal sample values, with linear interpolation between samples, gives the trajectory count $5KQ$. ■

Remark: This count is not the bitrate until quantisation, entropy coding, the carrier signal, and channel distortion are considered. The experiment reported in Section 8.6 retained $Q \in \{1, 3, 5, 9, 17\}$ temporal samples of each reconstructed complex coefficient trajectory, interpolating the real and imaginary parts linearly between them. That experiment therefore stores and interpolates $2NQ$ real values, which is $64Q$ at $N = 32$, and not $5KQ$. We report $5KQ$ alongside it only as a nominal signature-parameter budget proxy, indicating what an equivalent parameter-domain scheme would cost at the same temporal resolution. A genuine $5KQ$ parameter-rate result would require interpolating the fitted parameter trajectories instead, which we have not done.

7.3 Extrapolation and unseen-harmonic prediction

After fitting lower harmonics, the analytic map can be evaluated at unseen n.

Proposition 7.2 (First-order extrapolation error)**.** *For a local parameter error $\Delta\eta$ and the raw Jacobian $j_n(\eta) = \frac{\partial A_n}{\partial \eta}$,*

$$A_n\big(\hat{\eta}\big) - A_n(\eta) \;=\; j_n(\eta)\,\Delta\eta \;+\; O\big(\|\Delta\eta\|^2\big), \tag{7.3}$$

and hence

$$\big|A_n\big(\hat{\eta}\big) - A_n(\eta)\big| \;\le\; \|j_n(\eta)\|\,\|\Delta\eta\| \;+\; O\big(\|\Delta\eta\|^2\big). \tag{7.4}$$

Proof. Equation (7.3) follows from Taylor's theorem with the remainder applied to the C^2 map $\eta \mapsto A_n(\eta)$ at the point η with increment $\Delta\eta = \hat{\eta} - \eta$. Equation (7.4) follows from applying the Cauchy–Schwarz inequality to the linear term of (7.3), $\big|j_n\,\Delta\eta\big| \le \|j_n\|\,\|\Delta\eta\|$, and then taking moduli in (7.3) and applying the triangle inequality. ■

Remark: The derivative j_n contains factors n, $\log n$, and $e^{-\tau n}$ (Proposition 4.12). The factor $e^{-\tau n}$ shrinks the absolute extrapolation error at high n, but the relative error grows

because the target itself approaches zero. Hence we report the harmonic energy, the complex and magnitude errors, a phase-gauge-aligned error, the adjacent unseen harmonics, the four strongest-energy unseen harmonics, and a comparison against the zero predictor, whose complex NRMSE is exactly one (Lemma 8.2).

7.4 Structured denoising

Proposition 7.3 (Local linearised denoising ratio)**.** *At an interior true point η_0, let $y_R = A_R(\eta_0) + \varepsilon_R$ denote the realified noisy coefficient vector, where ε_R is the realification of $\varepsilon \sim \mathcal{CN}(0, \sigma^2 I)$ and $A_R = [\operatorname{Re} A_N;\ \operatorname{Im} A_N] : \mathbb{R}^{5K} \to \mathbb{R}^{2N}$ is the realification of A_N, and let $\hat{\eta}$ be the nonlinear least-squares estimate. To first order,*

$$\hat{\eta} - \eta_0 \ \approx\ J_R^{\dagger}\, \varepsilon_R, \tag{7.5}$$

the reconstructed coefficient perturbation is

$$A_R(\hat{\eta}) - A_R(\eta_0) \ \approx\ P_J\, \varepsilon_R, \qquad P_J \ =\ J_R J_R^{\dagger}, \tag{7.6}$$

and if $\operatorname{rank}(J_R) = d$*,*

$$E\|P_J \varepsilon_R\|_2^2 \ =\ \frac{\sigma^2}{2}\, d, \qquad E\|\varepsilon_R\|_2^2 \ =\ \frac{\sigma^2}{2}\,(2N), \tag{7.7}$$

so the ideal small-noise retained-noise ratio is

$$\frac{d}{2N} \ \approx\ \frac{5K}{2N}. \tag{7.8}$$

Proof. For (7.5), linearise the model around η_0: $A_R(\eta) \approx A_R(\eta_0) + J_R(\eta - \eta_0)$. The least-squares problem $\min_\eta \|y_R - A_R(\eta)\|^2$ becomes, to first order, $\min_\Delta \|\varepsilon_R - J_R\Delta\|^2$ whose minimum-norm solution is $\Delta = J_R^{\dagger}\varepsilon_R$ by the pseudoinverse characterisation of linear least squares [6].

For (7.6), applying the linearisation again, we find that $A_R(\hat{\eta}) - A_R(\eta_0) \approx J_R(\hat{\eta} - \eta_0) \approx J_R J_R^{\dagger}\varepsilon_R = P_J\varepsilon_R$, where P_J is the orthogonal projector onto $\operatorname{range}(J_R)$ [6].

For (7.7): since $\varepsilon \sim \mathcal{CN}(0, \sigma^2 I)$, each real coordinate of the realification has variance $\frac{\sigma^2}{2}$, because the real and imaginary parts of such a random variable are independent zero-mean Gaussian random variables each having variance equal to half of the complex Gaussian variance [11]. Thus, $E[\varepsilon_R \varepsilon_R^T] = (\frac{\sigma^2}{2}) I_{2N}$. Using the cyclic property of the trace and the symmetry and idempotency of P_J: $E\|P_J\varepsilon_R\|^2 = E\ \operatorname{tr}(P_J \varepsilon_R \varepsilon_R^T P_J^T) = \operatorname{tr}(P_J E[\varepsilon_R\varepsilon_R^T] P_J^T) = \frac{\sigma^2}{2} \operatorname{tr}(P_J P_J^T) = \frac{\sigma^2}{2} \operatorname{tr} P_J = \frac{\sigma^2}{2}\, d$, where the last equality uses the fact that the trace of an orthogonal projector equals its rank. The same computation holds for P_J replaced by I_{2N}, and its trace is $2N$.

For (7.8), divide the two expectations using $d = \operatorname{rank}\big(J_R\big) = 5K$ at a locally identifiable interior point (Definition 4.16). ■

7.5 Descriptor control, morphing, and classification

Brightness, tilt, peak location, energy, and flux depend deterministically on $\Sigma(t)$ (Section 6). Hence, they can be used to compare the dynamic characteristics of two musical notes once their loudness, carrier phase, and time origin have been fixed. For any two trajectories with the same gauge-fixed and component-matched form, the interpolation can be directly performed in the log coordinates for the positive parameters, along circular geodesics for φ, and in the complex Cartesian or magnitude-phase coordinates for c away from zeros. Moreover, the presence of collisions, the births and deaths of components, and different K values for the two trajectories require an explicit extension instead of an arbitrary one-to-one match.

The descriptors derived from the coefficients can further serve to discriminate between instrument classes.

8 Experimental Evaluation

In this section, we conducted an empirical evaluation of the developed model. Each subsection first outlines the protocols associated with an experiment, which is followed by the respective results. Section 8.1 describes the analysis pipeline used throughout. Rather than using the regularised path estimator of Section 5.4, we performed fits using the framewise estimator (4.7) with warm starts. This was done because the joint estimator of Section 5.4 introduces temporal regularisation that may smooth genuine rapid parameter changes, whereas the framewise estimator performs better as a validated benchmark.
All the programs used in this experimental evaluation can be found on this github repository: `https://github.com/swastikdash2309/styau-math-github-code.git`
The code can also be found on this drive: `https://drive.google.com/file/d/1GZD1q1Qbc6EzKstmktL2IcWP63KO8L6V/view?usp=sharing`

8.1 Protocol: estimator, datasets, and metrics

Estimator. We implemented the framewise variable-projection fit (4.7) on the admissible box $\tau_k \in \big[5\times10^{-4},\ 0.5\big]$, $\theta_k \in [0.05, 4]$. We fixed $K = 2$ for every real note, and used eight random starts on the first frame of a note along with one warm-started continuation start on every subsequent frame. For synthetic experiments, we used $K \in \{1, 2, 3\}$ together with a full fitted-order sensitivity grid.

Front end and datasets. The convention introduced in Section 7.1 was used to analyse real recordings. There was one constant f_0 per note, 17 Hann-windowed hetero-

dyne frames whose window length was at least 25 ms and four nominal periods, capped at 120 ms, and $N = 22$–32 harmonics under a Nyquist guard. We split each note's harmonics into a fitted prefix M (typically 17), an adjacent validation block (typically 6), and a far-upper block (typically 9). The dominant split used was $(N, M) = (32, 17)$. In this evaluation, we used two panels of the University of Iowa Musical Instrument Samples [4]. These included a development panel (36 notes, six instruments: bass clarinet, bass flute, B♭ clarinet, oboe, tenor trombone, tuba) and an independent replication panel (36 notes, six disjoint instruments: alto saxophone, bass trombone, B♭ trumpet, cello, soprano saxophone, violin). We drew six notes per instrument using a deterministic pitch-range-filtered rule, and no note was selected using model error. Waveform scores were used to cover the span between the first and last frame centres.

References for reconstruction. We evaluated reconstruction quality against three references: the original recording (waveform SNR and SI-SDR [12] of the model resynthesis against the recorded audio), the extracted harmonic path (the resynthesis of the measured coefficients themselves, labelled harmonic-reference SNR), and the measured coefficients (complex NRMSE of Definition 8.1).

The following error metric was used throughout.

Definition 8.1 (Complex NRMSE)**.** For a prediction $\hat{a}$ of a coefficient block $a \neq 0$, define the complex NRMSE:

$$\mathrm{NRMSE}\big(\hat{a}, a\big) \;=\; \frac{\|\hat{a} - a\|_2}{\|a\|_2}. \tag{8.1}$$

Lemma 8.2 (Zero-predictor calibration)**.** *For a zero predictor, the complex NRMSE for any nonzero coefficient block is equal to exactly one.*

Proof. For $\hat{a} = 0$ the complex NRMSE is $\frac{\|-a\|_2}{\|a\|_2} = \frac{\|a\|_2}{\|a\|_2} = 1$. ∎

Hence, an NRMSE near 1 is no better than predicting zero.

8.2 Experiment 1: Synthetic recovery: stationary fits and model order

Protocol. We generated random stationary signatures with $K \in \{1, 2, 3\}$ well-separated components and $N = 48$ harmonics. We gave the estimator the first $M \in \{8, 12, 16, 24\}$ noisy coefficients at noiseless, 40 dB, and 20 dB conditions, with 12 trials for each setting. Then, performance was measured on the remaining $N - M$ unseen harmonics using complex NRMSE. We also ran a 54-fit test at $M = 16$ and 40 dB to investigate the effect of fitting the wrong model order K.

All 432 stationary fits produced finite outputs and reported successful execution. In the noiseless condition, median held-out complex NRMSE was 1.65×10^{-13} for $K = 1$,

6.40×10^{-13} for $K = 2$, and 4.21×10^{-12} for $K = 3$. With noise present, error rose as K increased and fell as more harmonics were fitted (Figure 1).

Table 1. Held-out complex NRMSE by coefficient SNR and model order.

Coefficient SNR	$K = 1$ held-out NRMSE	$K = 2$ held-out NRMSE	$K = 3$ held-out NRMSE
noiseless	1.65×10^{-13}	6.40×10^{-13}	4.21×10^{-12}
40 dB	0.0116	0.0354	0.0799
20 dB	0.1185	0.3945	0.5683

At 40 dB, the median $K = 1$ parameter errors were 7.32×10^{-4} in τ, 4.62×10^{-4} rad in φ, 0.00569 in θ, and 0.0116 relative in c. The matching $K = 2$ errors were 0.0122, 0.00201 rad, 0.0969, and 0.100. For $K = 3$, they were 0.0282, 0.0132 rad, 0.214, and 0.378. As expected, stability declined when the number of components and nonlinear degrees of freedom increased (see Section 4.2).

In the 40 dB order grid, held-out error selected the diagonal structure. Median held-out NRMSE for true/fitted $K = (1, 1)$, $(2, 2)$, and $(3, 3)$ was 0.0172, 0.0189, and 0.0320. When a true $K = 3$ signal was underfit with $K = 1$, the median rose to 0.707. Fitting $K = 2$ gave 0.147. Mild overfitting caused less damage in this small grid, but it took longer and did not improve held-out error.

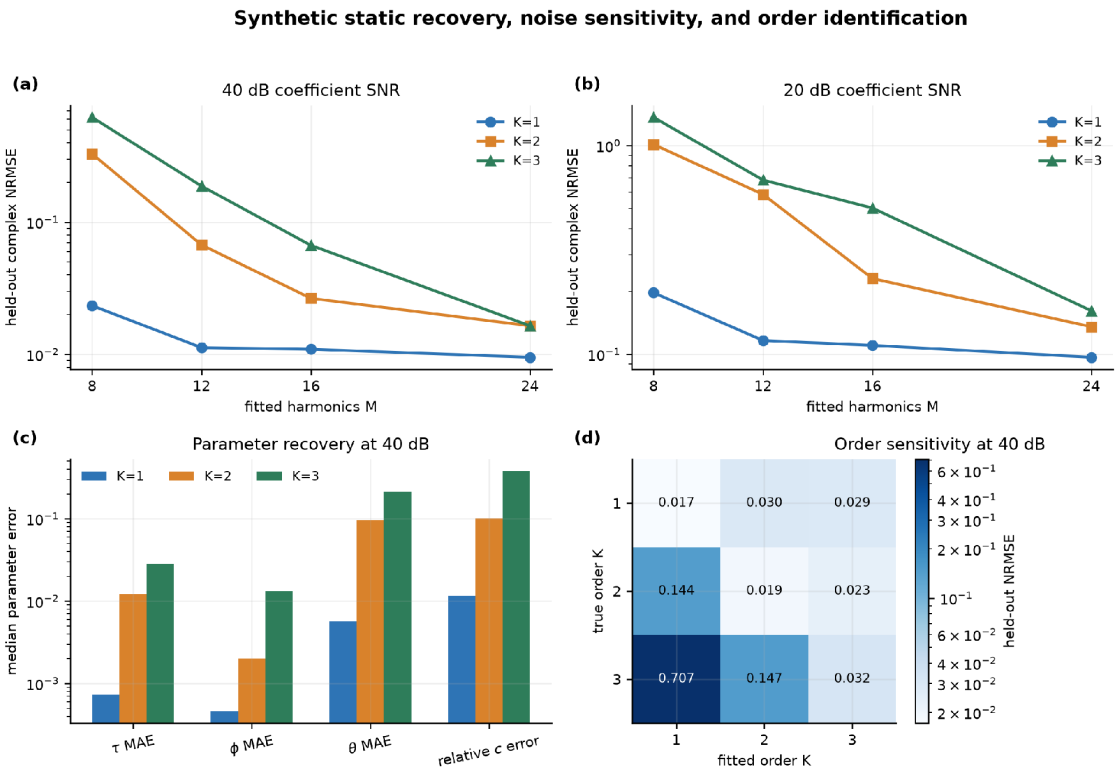


Figure 1. Panels (a) and (b) plot held-out complex NRMSE against the number of fitted harmonics at 40 and 20 dB. Panel (c) reports parameter errors at 40 dB. Panel (d) shows the held-out error matrix for true order versus fitted order.

8.3 Experiment 2: Dynamic trajectory recovery and conditioning

Protocol. We generated a $K = 2$ time-varying signature over 32 harmonics and 21 frames. At each frame, the estimator fitted the first 16 noisy harmonics with the warm-started framewise method. We tested eight scenarios, which included seven correctly specified cases and one misspecified case with a third component. Six trials were conducted at 40 dB and 20 dB SNR. Parameter errors were measured after matching the recovered components to the true ones in each frame, while a separate conditioning test tracked how the Jacobian becomes unstable as the two components move closer together.

Across the seven correctly specified dynamic scenarios at 40 dB, pooled median per-frame errors were 0.0105 in τ, 0.00309 rad in φ, 0.0433 in θ, and 0.0588 relative in c (Figure 2). At 20 dB, the values increased to 0.0954, 0.0348 rad, 0.3737, and 0.3698. Near collisions and in low-amplitude frames, reconstruction of the coefficients remained much more stable than recovery of the individual coordinates.

Table 2. Dynamic scenarios: all-coefficient and held-out complex NRMSE at 40 and 20 dB.

Scenario	All NRMSE, 40 dB	Held-out NRMSE, 40 dB	All NRMSE, 20 dB	Held-out NRMSE, 20 dB
stationary	0.00677	0.0119	0.0656	0.113
gain only	0.00691	0.0123	0.0864	0.174
slow spectral shape	0.00566	0.0108	0.0604	0.123
fast attack	0.00646	0.0124	0.0613	0.118
coherent phase motion	0.00743	0.0120	0.0778	0.121
component crossing	0.0124	0.0471	0.111	0.412
near collision	0.0187	0.0796	0.239	1.027
misspecified residual	0.0713	0.0778	0.254	0.275

The near-collision and misspecified cases mark the practical boundary of the inverse model. When phase separation decreased from 1 rad to 0.01 rad, the condition number of the realified Jacobian increased from 97.5 to 1.87×10^8. At exact coincidence, its rank fell from 10 to 6. This confirmed that the counting requirement $2N \geq 5K$ is necessary but not sufficient (Proposition 4.15, Definition 4.16); local recoverability also depends on component separation and amplitude.

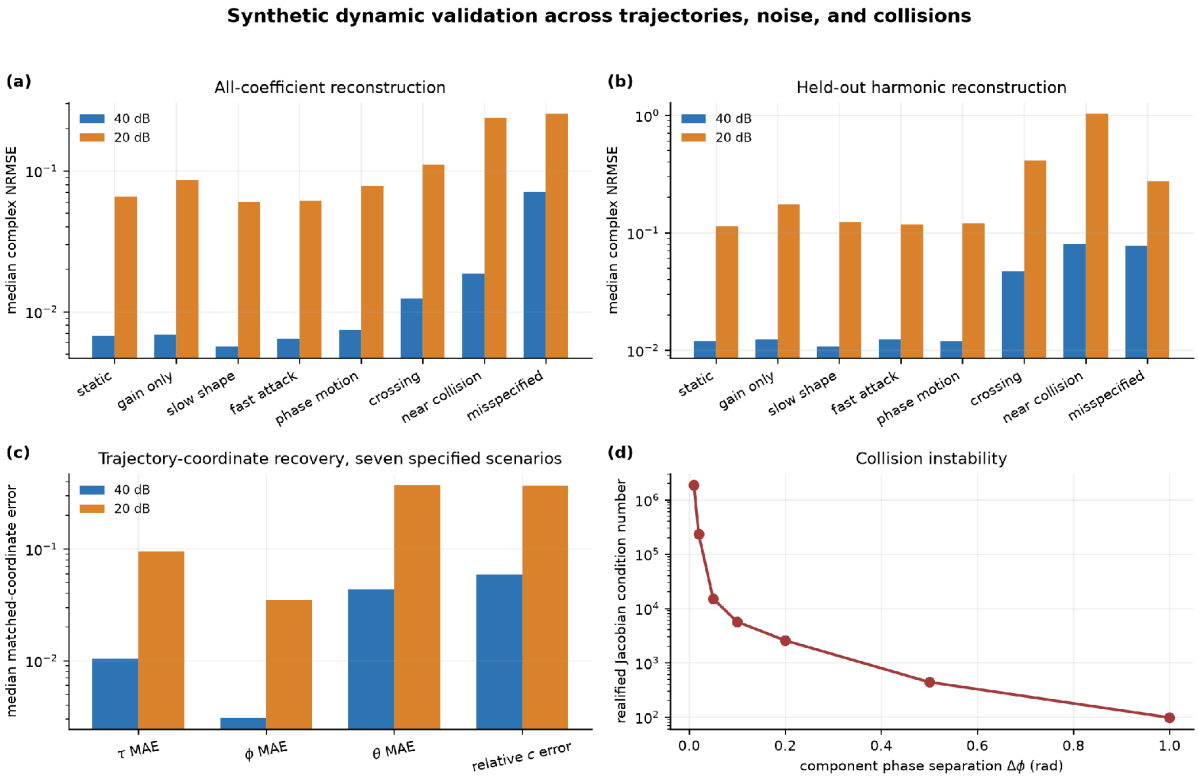


Figure 2. Panels (a) and (b) report all-coefficient and held-out complex NRMSE for eight scenarios. Panel (c) shows matched trajectory-coordinate error over the seven correctly specified scenarios. Panel (d) shows the Jacobian becoming ill-conditioned as components collide.

We ran the fast-attack scenario end-to-end with the same generator, seeds, and estimator, comparing the recovered parameter paths frame by frame against the ground truth (Figure 3). This was done to make the trajectory claim concrete. The dominant component's depth $\tau_1(t)$, type $\theta_1(t)$, and gain $\left|c_1(t)\right|$ were tracked through the 50 ms attack transient and the sustain drift, and in the sustain region the median relative error in τ_1 was 2.4%. The first analysis frame's window straddled the fastest part of the attack, and the deep component $\tau_2(t)$'s contribution decayed towards the noise floor. But both of these showed visible deviations in the two failure modes anticipated by the window-bias law (5.11) and by the identifiability degradation of Section 4.4. The same run was carried to the waveform level in panel (e): the true synthetic waveform, the oracle resynthesis from the extracted coefficients, and the resynthesis from the fitted signature trajectory were overlaid. The signature reconstruction was within 0.1 dB of the 40.6 dB raw-coefficient oracle as it was 40.7 dB SNR against the true waveform. This was because the parametric law regularizes the per-frame coefficient noise.

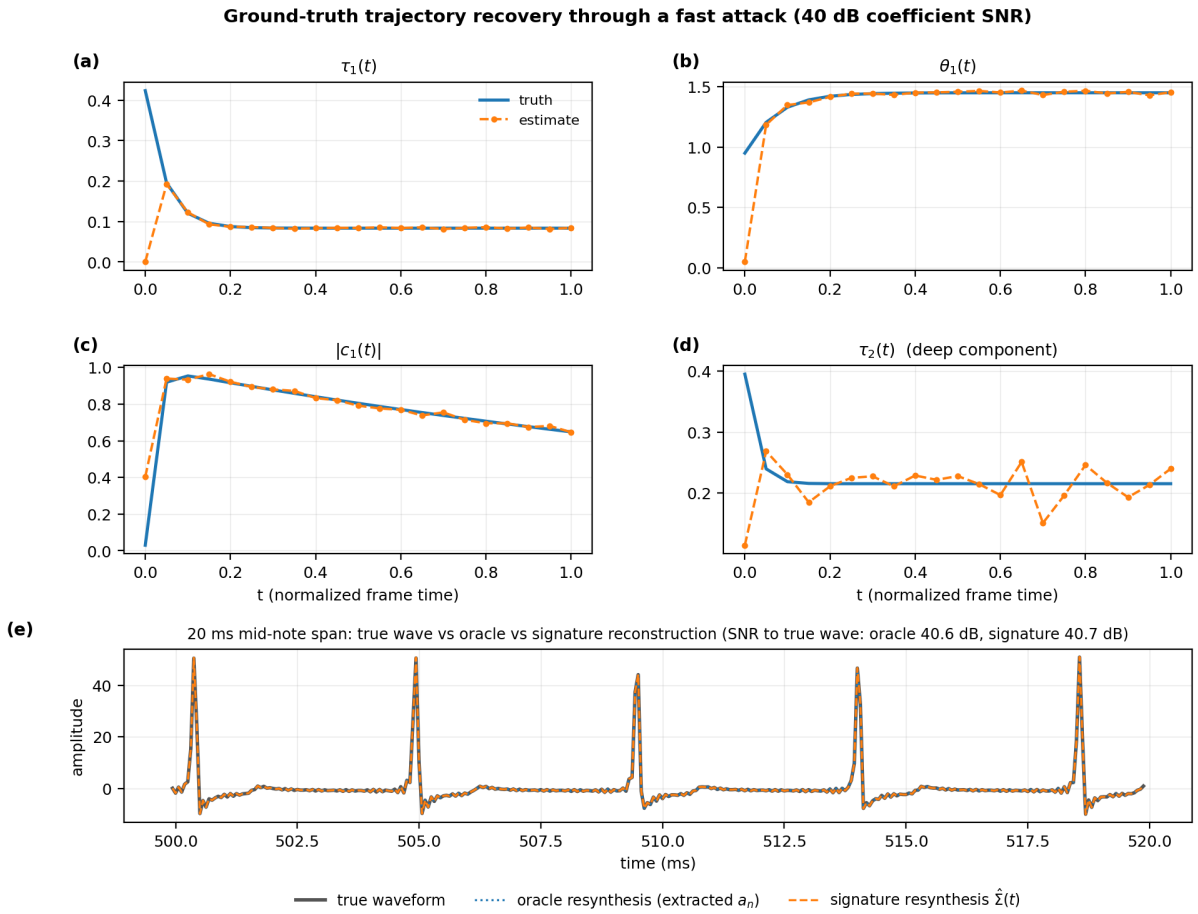


Figure 3. Panels (a)–(d) show recovered parameter trajectories (dashed) against truth (solid) for the dominant component's depth, type, and gain, and for the deep component's depth. The deep component's jitter reflects its fading amplitude, and the first-frame deviation demonstrates window bias during the attack. Panel (e) shows a 20 ms mid-note span of the true waveform, and the resynthesis from both extracted coefficients and from the fitted singularity signature.

8.4 Experiment 3: Real tones: coefficient recovery and sound reconstruction

Protocol. We processed each note in the two Iowa datasets using the analysis pipeline outlined in Section 8.1, and fitted frame-by-frame with $K = 2$. We measured reconstruction quality in three ways, including a complex NRMSE against the extracted harmonic coefficients, SNR against the extracted harmonic signal, and recording SNR and SI-SDR against the original audio. One representative note from each panel was then chosen as the note whose coefficient NRMSE was closest to that panel's median.

We performed 72 real-note runs. The pooled median all-coefficient NRMSE was 0.2530, with a 95% dataset-instrument cluster interval of 0.181–0.322 (Figure 4). Median SNR relative to the extracted harmonic path was 12.75 dB (10.63–15.33 dB). Median SNR relative to the original recording was 5.27 dB (3.84–8.27 dB), and median SI-SDR

was 3.76 dB (1.57–7.56 dB). The gap between harmonic-reference and recording scores shows both residual channels and front-end limits as well as reconstruction error.

Table 3. Real-note panel results.

Panel	Coefficient NRMSE	Harmonic SNR	Recording SNR	SI-SDR
Iowa development	0.2375	12.77 dB	5.27 dB	3.76 dB
Iowa replication	0.2550	12.75 dB	5.29 dB	3.78 dB

Across all notes, median harmonic-reference SNR was 12.78 dB in the early span, 13.57 dB in the middle span, and 15.78 dB in the late span. These values summarize three equal frame spans. They are not attack, sustain, and release boundaries identified by perception.

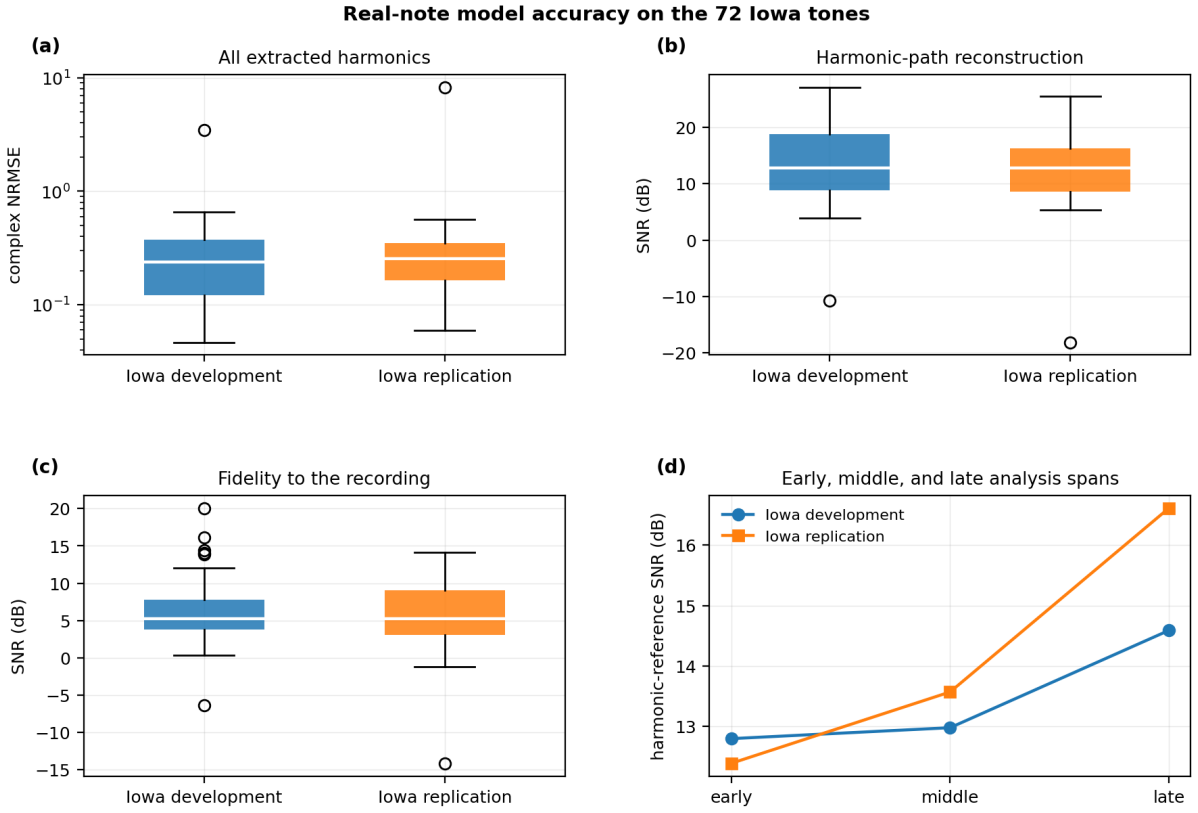


Figure 4. Panels (a)–(c) show note-level distributions for all-coefficient error, harmonic-path SNR, and recording SNR. Panel (d) shows median harmonic-reference SNR over the early, middle, and late analysis spans for each panel.

Figure 5 uses a fixed selection rule for one note from each panel. This was the note whose all-coefficient NRMSE was nearest to that panel's median. The Iowa tenor-trombone example had NRMSE 0.2577, harmonic-reference SNR 11.85 dB, and recording SNR 5.77 dB. The Iowa alto-saxophone example had NRMSE 0.2616, harmonic-reference SNR 12.56 dB, and recording SNR 9.60 dB.

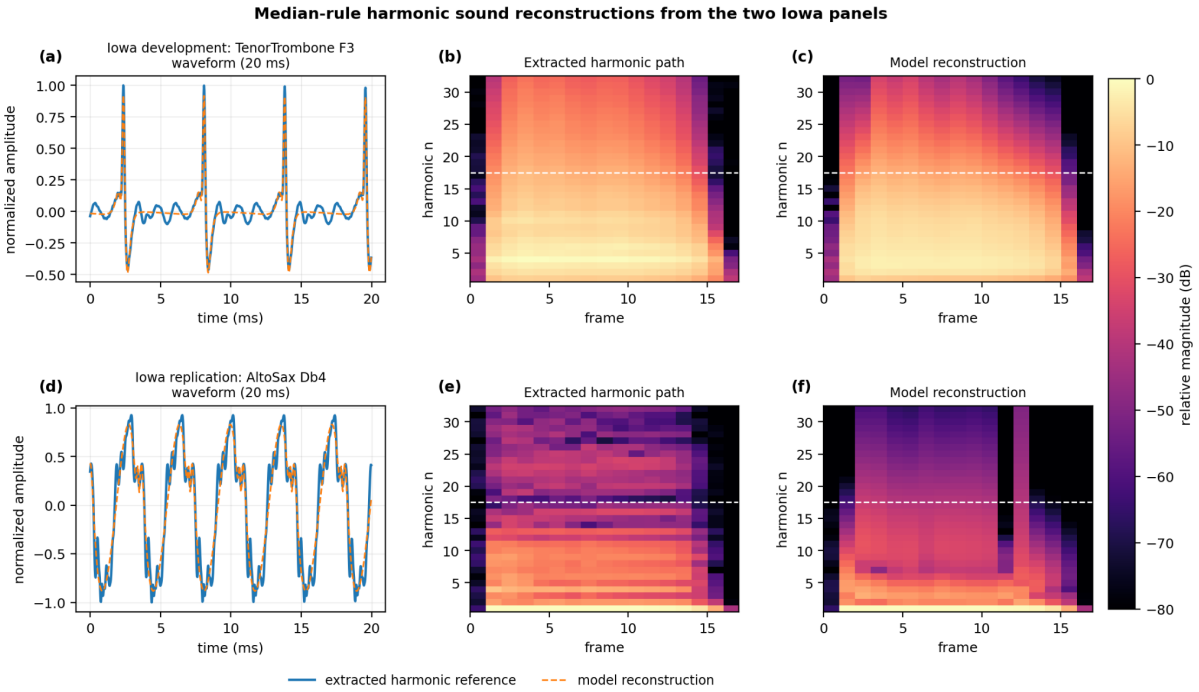


Figure 5. The left column overlays a 20 ms extracted harmonic waveform with its model reconstruction. The centre and right columns show the reference and reconstructed harmonic-magnitude trajectories. Dashed lines indicate the boundary of the fitted low-harmonic prefix.

8.5 Experiment 4: held-out Fourier-coefficient prediction

Protocol. In this experiment, we tested whether a singularity signature fitted to lower Fourier coefficients can predict unseen higher coefficients. In the stationary synthetic tests, the model was fitted using the first M coefficients and then evaluated on the remaining $N - M$ coefficients. To examine the error across harmonic numbers, the experiment was re-run at the $K = 2$, $M = 24$, 40 dB setting with the same generator, seeds, and estimator.

Results. In this experiment, the prediction accuracy grew steadily with the size of the fitted block. At 40 dB, fitting the largest prefix ($M = 24$) and predicting the 24 unseen coefficients gave median complex NRMSE 0.0095 for $K = 1$ and 0.0165 for $K = 2$ and $K = 3$. Per harmonic, the median relative error was 0.40% inside the fitted prefix and 1.5% on the predicted harmonics. This stayed below a few percent out to $n = 48$ (Figure 6). Figure 6(a) shows a representative fit where the predicted magnitudes continue the true envelope through twenty-four harmonics the estimator never observed. The same behaviour extended to moving signatures. In the dynamic scenarios given in Table 2, the held-out blocks of the seven correctly specified trajectories were predicted with median NRMSE 0.011–0.080 at 40 dB.

On real notes, the extracted coefficient energy was concentrated almost entirely inside the fitted prefix, so held-out prediction was quantitative only within the model class:

median coefficient-energy fractions were 99.951% in the fitted prefix, 0.0322% in the validation block, and 0.0094% in the far-upper block.

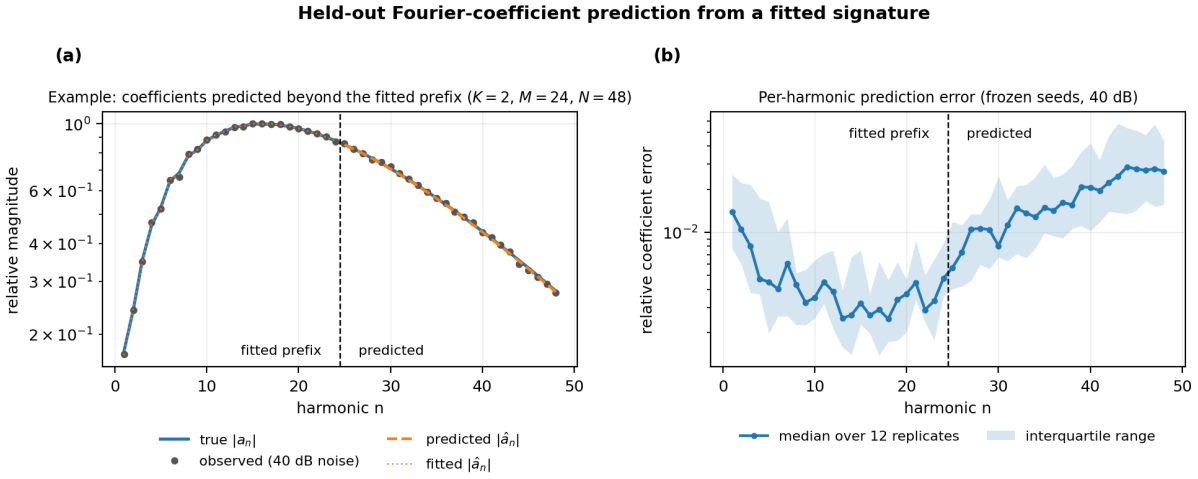


Figure 6. Panel (a): this is a representative example in which the signature is fitted on the first 24 noisy coefficients (left of the dashed line), and its analytic continuation predicts the next 24, tracking the true envelope. Panel (b): this shows the per-harmonic relative prediction error, median and interquartile range over the 12 frozen replicates. The median is 0.40% inside the fitted prefix and 1.5% on the predicted block.

8.6 Experiment 5: Robustness: rate sensitivity, classification, and diagnostics

Protocol. We performed three additional tests that used the fitted real-note trajectories from Experiment 3. We implemented rate sensitivity to test reconstruction using only $Q \in \{1, 3, 5, 9, 17\}$ temporal samples of each reconstructed complex coefficient trajectory, linearly interpolating the real and imaginary parts between them. Also, we performed instrument classification using a three-fold nearest-centroid classifier built from summary features of the first 12 harmonics, including their average behaviour, variation over time, and early–middle–late differences. Finally, diagnostic tests recorded parameter-bound activity, runtime, and sensitivity to deliberate carrier-frequency offsets.

Temporal-knot sensitivity showed that all-coefficient error fell monotonically as temporal resolution increased (Figure 7). Median NRMSE at 1, 3, 5, 9, and 17 knots was 1.221, 0.914, 0.839, 0.635, and 0.253. The values actually stored and interpolated are the $2NQ$ real coefficient knots, that is 64, 192, 320, 576, and 1088 at $N = 32$; the budgets of 10, 30, 50, 90, and 170 quoted on the horizontal axis of Figure 7 are the nominal signature-parameter proxy $5KQ$ of Proposition 7.1, not the storage used here. Because this experiment interpolated reconstructed coefficient trajectories rather than fitted parameter trajectories, and did not quantise or encode the parameters, it supports only a within-model statement about sensitivity to rate.

Three-fold macro instrument-classification accuracy was 0.583 for Iowa development and 0.500 for Iowa replication. The corresponding nominal chance levels were $\frac{1}{6}$ for both panels. These small exploratory folds suggest that the dynamic summaries contain instrument information, but they do not establish a perceptual timbre distance or broad classification performance.

Activity at parameter bounds was substantial. Across real component coordinates, 63.3% of τ values reached the upper bound, 31.1% of θ values reached the lower bound, and 9.9% of θ values reached the upper bound. Median runtime was 0.284 s per note, with a cluster interval of 0.246–0.326 s. The convergence flag for the winning optimiser run was not available. Accordingly, 'completed' means that no exception occurred and the audited output was finite. It does not certify a global minimum (see (4.7)).

A constructed carrier-offset test produced essentially zero coefficient error through 40 cents because framewise changes in $\varphi(t)$ absorbed the common phase drift. This confirmed the warning in Proposition 5.4: without a phase convention fixed from outside the model, phase motion cannot be interpreted uniquely as timbre motion.

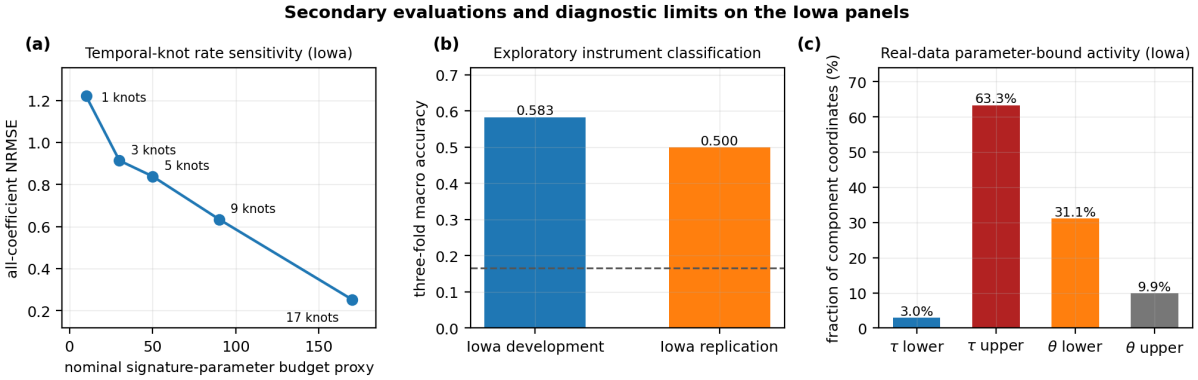


Figure 7. Panel (a) reports sensitivity to the temporal-knot rate, plotted against the nominal signature-parameter budget proxy $5KQ$; the coefficient knots actually interpolated number $2NQ$. Panel (b) reports exploratory instrument-classification accuracy against the $\frac{1}{6}$ chance level. Panel (c) shows how often real-data parameters reached their bounds.

9 Conclusion

This paper establishes dynamic complex singularity signatures as a unified analytic representation of evolving harmonic spectra. A finite mixture whose singularity depth, angular position, type, and complex amplitude vary with time generates the local sequence of harmonic coefficients. From this coefficient map, we derive the generating function and its singular data, a signature space defined up to permutation, variable projection recovery, and recurrence and matrix pencil initialisers for integer types. We also derive the exact Jacobian and Fisher information, infinite-data identifiability, finite-data local rank conditions, a windowed dynamic law with a fixed phase gauge, an explicit reconstruction

error chain, and a descriptor calculus for brightness, spectral shape, temporal evolution, and spectral flux.

The experiments test the theory and provide validation. Recovery is shown to be exact to machine precision, supporting Theorem 4.14's prediction of exact recovery. Moreover, Proposition 4.15 stated that parameter counting is insufficient, and this was validated by the rise of the Jacobian's condition number from 97.5 to 1.87×10^8 as components merged. By fitting a signature to 24 coefficients, it was able to predict previously unobserved 24 coefficients at 1.5% median error, which per-harmonic models cannot attempt. Across 72 tones from two separate Iowa panels, the model retained 93.6% of the coefficient energy using only 170 values instead of 1,088, with similar results on the second panel and no instrument-specific tuning.

However, this research is also subject to several limitations. The model describes only the phase-coherent harmonic component of quasi-harmonic tones, and does not fully cover timbre. For example, the model fails to capture inharmonic, stochastic, or reverberant sound components. It may also poorly represent pitch-dependent formants, and reconstructing the harmonic component does not reproduce the full recording. Furthermore, the recovery guarantees provided are local and may fail because of local minima, phase ambiguity, component collisions, or incorrect model order. Also, even if the reconstructed coefficients are accurate, parameters may become difficult to identify when components are closely spaced or weak. Limitations are present in the empirical evaluation as well, because complete sound reconstruction requires residual and inharmonic channels, while the current model reconstructs only the harmonic component and the framewise estimator does not enforce temporal smoothness across the recovered trajectories.

Future work will include incorporating residual and inharmonic sound components, validating the joint temporal estimator, and testing perceptual relevance through blinded listening studies.

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